7

Natural Language Processing

7.1 What is Natural Language Processing?

Natural language processing is the manipulation or understanding of text or speech by a computer software. As humans interact, they understand each other views, and respond with an appropriate answer; similarly, NLP systems also interact, understand and respond to queries. The contrast between Natural Language and Computer Language can be understood from **Table 7.1**

Natural language processing, a field of artificial intelligence, studies interactions between computers and human (natural) languages. It applies **machine learning** algorithms to **text** and **speech**. For example, NLP can be used to create intelligent systems for **speech recognition, machine translation, question answering, spam detection, document summarization, predictive typing, named entity recognition** and **autocomplete, to name a few.**

Morphology studies the formation of words from stems, prefixes and suffixes.

These days, smartphones and computers support so many applications that all modern operating systems have a built-in speech recognition module. For example,

Cortana, the **virtual assistant** of Microsoft Windows, recognizes human **voice** to set up reminders, open apps, send emails, search information, play games, get latest updates on weather, etc.

Siri, the virtual assistant of the Apple Inc.'s iOS, watchOS, macOS, HomePod and tvOS helps users perform their tasks with **voice commands** like starting a call, texting someone, sending an email, clicking a picture, setting a timer, opening an app, setting an alarm, using navigation and so on.

TABLE 7.1 Differences between Natural language and Computer Language

Natural Language	Computer Language
Natural language has a very large vocabulary.	Computer language has a very limited vocabulary.
Natural language is easily understood by humans.	Computer language is easily understood by machines.
Natural language is ambiguous in nature.	Computer language is unambiguous.

Gmail, the most popular email service developed by Google, uses NLP to detect and filter spam emails.

The main reason behind the rise of NLP can be simply summarized in two points.

1. **Large volumes of textual data:** Natural language processing helps computers communicate with humans in their own language. NLP empowers computers to read text, hear speech, interpret it, measure sentiment and determine which parts are important.
   Today's machines are supposed to interact with humans in a consistent, unbiased way without getting tired. With increasing use of machines in our day-to-day lives (from medical records to social media), enormous amount of unstructured data is being generated.
2. **Structuring the unstructured data:** Human language is complex and diverse as humans express the same thing in infinite ways both in written and spoken form.

You will be surprised to know that there are 6500 human languages, out of which 121 are frequently spoken. Every language has its unique set of grammar and syntax rules, terms and slang. Moreover, while writ-

ing, we often misspell or abbreviate words, or omit punctuation. Similarly, while speaking, we pronounce different words differently (in regional accents). Some people mumble, stutter and borrow terms from other languages which further increases the complexity of processing natural languages.

Techniques like deep, supervised and unsupervised learning are widely used for modeling human language to resolve ambiguity in language.

How are NLP Systems useful?

NLP systems break down text/speech in human language into shorter, elemental pieces, to understand relationships that exist between the pieces for interpreting its meaning. Such systems are then used for the following tasks.

1. **Generate a summary of the document, generate keywords for efficient searching and indexing.**
2. Accurately capture the meaning and themes in text and use it for optimization and forecasting.
3. Automatically extract structured information from textual sources.
4. **Perform sentiment analysis to i**dentify user's opinion within large amounts of text.
5. Speech-to-text and text-to-speech conversion to transform voice commands into written text, and vice versa.
6. Automatic translation of text or speech from one language to another.

Spam emails are those emails that are sent in bulk to recipients without their permission. They are often sent for commercial purposes.

7.2 Everyday NLP Examples

Some widely used NLP applications in our everyday lives include the following:

Just check the spam folder in your email, it has already separated some emails with common words as spam. Spam filtering uses NLP to compare words in spam email to those present in valid emails to identify junk mail (refer **Fig. 7.1**).

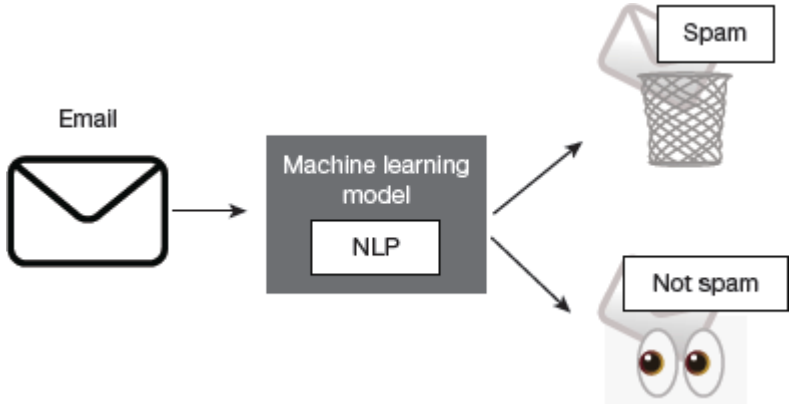


FIGURE 7.1 Classifying Email as Spam or Not a Spam

Whenever you miss a phone call, you get an automatic transcript of the voicemail in your email inbox or smartphone app. This is possible due to NLP's capability to perform speech-to-text conversion.

NLP is also the underlying technology when we do a voice search.

Speech recognition is used for converting spoken words into text. It is used in applications where the user gives voice commands to smart-phones, voice biometrics device, home automation systems or dictates in software like Microsoft Word, etc. (refer **Fig. 7.2**).



Credit: Text Recognition

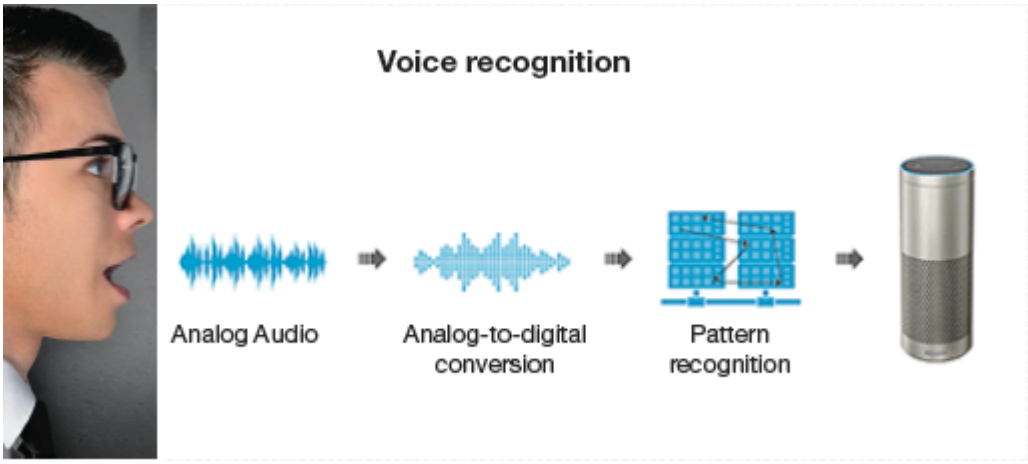
FIGURE 7.2 Performing Commands using Speech Recognition

In these systems, NLP is used to give voice commands to devices to perform a particular task (ex, play a song or search information about a topic on a search engine).

NLP systems are being used by companies to understand what their customers think and say about them on social media, emails, feedbacks and reviews.

Companies perform sentiment analysis (also known as **opinion mining) to assess** the attitude, behaviour and emotional state of customers. For this, NLP assigns values to the text (positive, negative or natural) and also identifies the mood of the context (happy, sad, angry, etc.) in which the customer has written that feedback/review.

Did you ever think what happens when we speak to our device? Actually, the microphones of the device accept our voice as input and plot the graphs of sound frequencies. Every sound has a unique frequency graph. This fact helps NLP to recognize each sound and compose an individual's words and sentences (refer **Fig. 7.3**).



Credit: fuzzbones / 123RF

FIGURE 7.3 Voice Recognition

Chatbots are NLP, AI-based systems that are specifically designed to interact with humans through text or speech. They are extensively used these days to offer 24 X 7 assistance to customers by handling their queries. Chatbots actively learn from each interaction to gain better understanding of user's intent so that they can be used by companies to au-

tomate repetitive and time-consuming tasks. This allows employees to focus on more fulfilling tasks.

Spelling correction is a very important tool provided by Microsoft Corporation in its application software like MS-Word and PowerPoint. The NLP program embedded in this software automatically checks the spelling in the text. If the spelling is incorrect, it notifies the user and also gives suggestion on what could be the correct spelling.

Topic classification uses NLP to organize unstructured text into categories. For example, companies use NLP to analyse hundreds of open-ended responses to surveys to gain insights from customer feedback. Topic classifier tags all keywords in seconds and then automatically routes them to the right person for speedy response.

NLP is used to translate text/speech written in one language to another; for e**xample:** Google Translator.

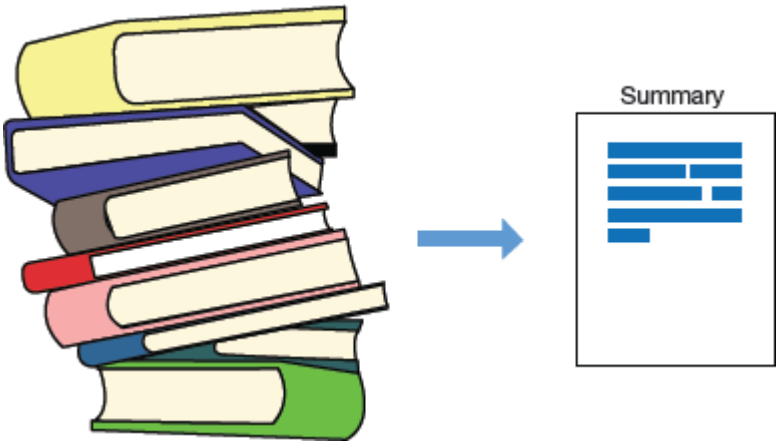
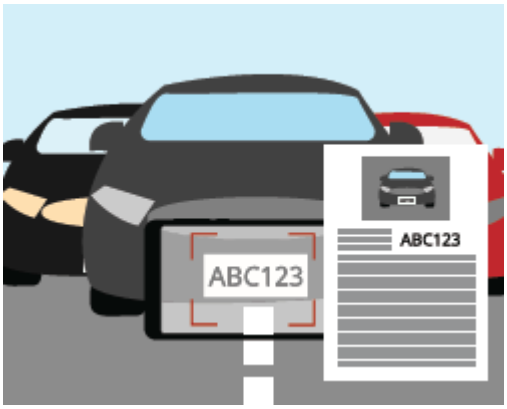


FIGURE 7.4 Text Summarization



Credit: Piscine26 / Shutterstock

FIGURE 7.5 Text Recognition

Text Recognition (in an image or video): These days, cameras are installed at major crossings. Have you ever noticed that if someone skips stopping during a red signal then automatically a challan comes to his/her house in a day or two? Did you try to think about the technology being used? Basically, the camera clicks the image of the vehicle along with its number plate (refer **Fig. 7.5**). NLP tool is then used to extract number from the number plate. Obviously, once we get the exact vehicle number, we can easily retrieve information about the owner such as name, age, address, phone number, date of purchase, registration number, etc. In this application, NLP is being used to recognize text from an image.

Text Summarization: NLP tools can easily read, understand and summarize long paragraphs of text or even full article into a shorter narrative without changing its inherent meaning (refer **Fig. 7.4**). Summarization is

very helpful in various applications. For example, researchers need to read several papers and articles published before finalizing their work. It is a very tedious job to read those documents line-by-line and make notes. A summarization tool is of great essence in this domain.

Lawyers and judges have to read through large volumes of documents to develop an understanding of the ongoing case. Even in this case, the summarization tool can read the case files to create a short abstract that can be used by legal practitioners, thereby saving their time and expediting the pending cases resolution. Text summarization is done in two ways.

First, key phrases are extracted from the document and combined to form a summary. This is known as **extraction-based summarization.**

The second option is **abstraction-based summarization** in which the source document is shortened. We create new sentences from the original content in this step. This is in contrast to extraction-based technique, in which we only utilized the phrases that were present. It is possible that the phrases formed by abstractive summarization are not present in the original text. Therefore, although abstraction models are more complex to build, they are more efficient than extraction.

The process of constructing a concise, cohesive and fluent summary of a lengthier text document, which includes highlighting the text's important points, is known as text summarization.

Information Extraction: Information extraction is the process of extracting information from unstructured textual sources to identify entities so that they can be efficiently stored in a database. Real-world example of information extraction is using email extracts to add relevant in your Calendar.

Information extraction extracts relevant information from unstructured text data and presents it in a structured format.

Information extraction techniques are widely used in healthcare, judiciary and to understand social media interactions and streams, online news, government documents, corporate reports and more to automate certain tasks (like smart content classification); integrate search, management and delivery tasks; identify patterns and trends to uncover hidden relationships. For example, go through the cricket news article given below.

Indian captain Virat Kohli was dismissed cheaply for just 2 in Wellington on Friday by debutant Kyle Jamieson extending a rare lull in the batsman's stellar career. Throughout the ongoing New Zealand tour, Kohli has managed to score just a single fifty across 8 innings in all 3 international formats.

From this unstructured text, we can extract the following information.

- Country – India, Captain – Virat Kohli
- Batsman – Virat Kohli, Runs – 2
- Bowler – Kyle Jamieson
- Match venue – Wellington
- Match series – New Zealand
- Series highlight – single fifty, 8 innings, 3 formats

Structured data helps us to take full advantage of powerful query tools like SQL for further analysis.

7.3 Chatbot

Chatbot (refer **Fig. 7.6**) is an artificial intelligence (AI) program that simulates conversation (or a chat) with a user in natural language through messaging applications, websites, mobile applications or by phone. You must have seen that many websites offer a real-time chat service that helps you to find relevant information on that website. Such an application provides a pleasant user experience and faster and simpler customer service interaction. At the same time, it also allows companies to improve their operational efficiency by reducing the typical cost of customer service. Employees that were dedicatedly providing customer support can

now be utilized to do more fruitful jobs. Companies are now able to save thousands of rupees per month by deploying a chatbot instead of a human employee dedicatedly working to provide customer support.

The best part of a chatbot is that it allows users to pick up a conversation from the point where they last left. This has been possible because chatbots have a memory that implicitly learns about a use and stores key information about it for future use. The advantages of chatbots are summarized in **Fig. 7.7**

Users can query about multiple things one after the other without annoying the information provider.



Credit: Tero Vesalainen / Shutterstock

FIGURE 7.6 Chatbot

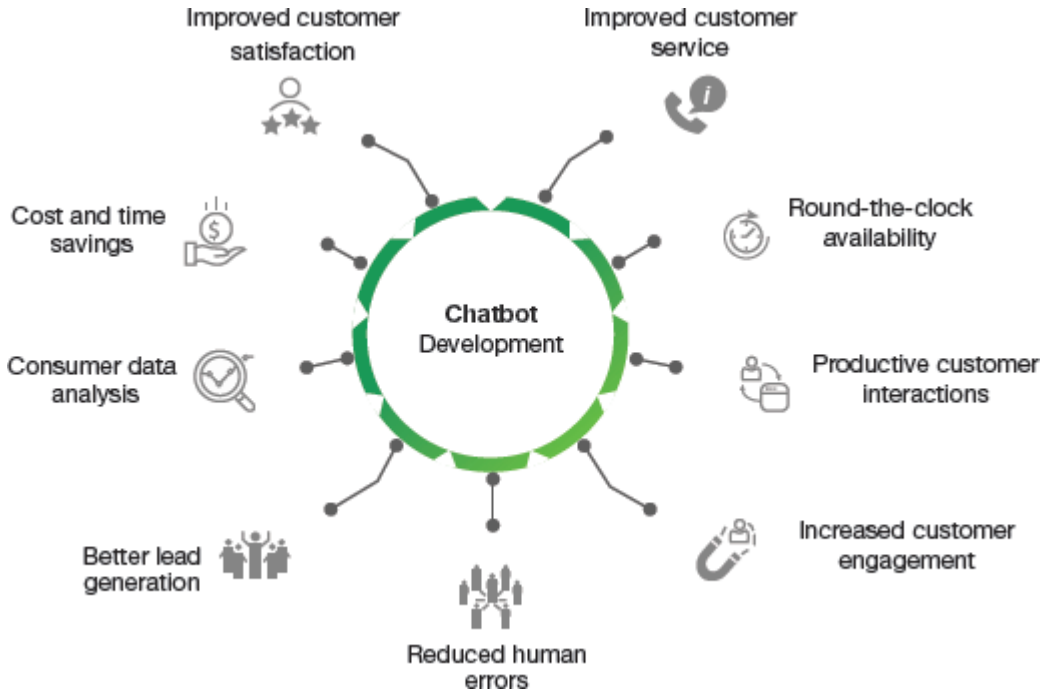


FIGURE 7.7 Advantages of Chatbots

7.3.1 Open and Closed Chatbots

A chatbot can be categorized as open or closed depending on its usage. While open chatbots use ML techniques to learn from their interactions with users, closed chatbots, on the other hand, execute a specific conversation flow.

Thus, open chatbots support a more natural way of conversing with humans. But their success very depends on how well they have been trained. Closed chatbots comparatively have a lower risk of providing poor user experience.

7.3.2 Uses of Chatbots

1. **Customer service: Chatbots can be programmed to co**llect registration data, provide information, answer frequently asked questions, etc. This gives a pleasant experience to customers. Users can now interact in a simpler and faster way.
2. **Online purchases:** Chatbots assist users while shopping online to select a product of their utility and then make the payment easily. They provide information, solve problems faced by users during shopping and perform searches to identify similar products.
3. **Sending information and news about latest updates in products or about discount offers** automatically to customers and potential buyers of the product.
4. **Collect Information: C**hatbots can be designed to collect some useful information about users, such as email, phone, etc. This information can be filtered (to select most appropriate ones in the current context) and then be forwarded to sales representatives for a follow-up conversation.
5. **Make Appointments:** Chatbots can automatically make appointments by displaying availability on a calendar and allowing the user to select a suitable date.
   Healthcare websites deploy chatbots in their websites to provide patient information. A patient can enter his symptoms. The NLP-based chatbot will check those symptoms with known ailments. After analysing the basic issues, the chatbot can then *assist patients in booking an appointment* (refer **Fig. 7.8**), *finding a doctor and more.*
   Chatbots also help users to *find an appropriate insurance policy*. For this, the chatbot collects useful information from users and then tailor products according to the consumer's needs. These chatbots not only help users find a suitable policy but also complete the payment in a hassle-free way, file for a claim, renew policies, check their coverages, track status of policy/claim, to name a few.

Children of this era, known as Gen Z, are dealing with more mental health, body image, and loneliness issues than others. These *children can use chatbots that can gauge their emotional state based on conversational context and respond with compassion.*

6. **Builds trust and improves customer engagement:** Chatbots help companies to keep their customers satisfied, well-informed as they are assisted in everything they do. A chatbot instantly responds 24 X 7 **to** provide immediate answers to simultaneous users at once.

7.3.3 How a Chatbot Works?

A chatbot works on a request-response model. When a user requests information, the *chatbot* analyses the request to **identify what information the user exactly wants.**

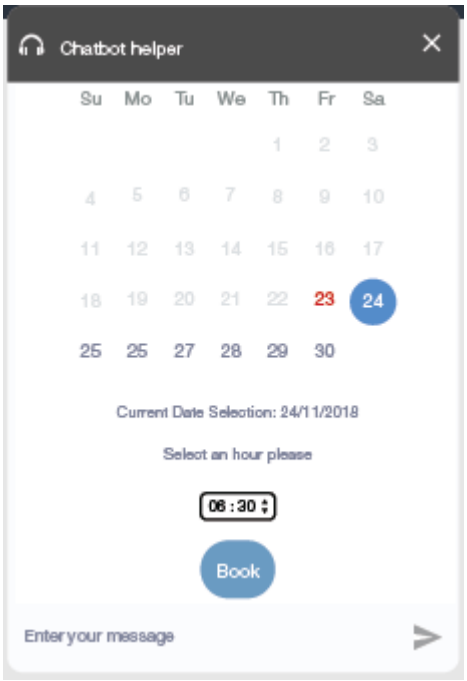
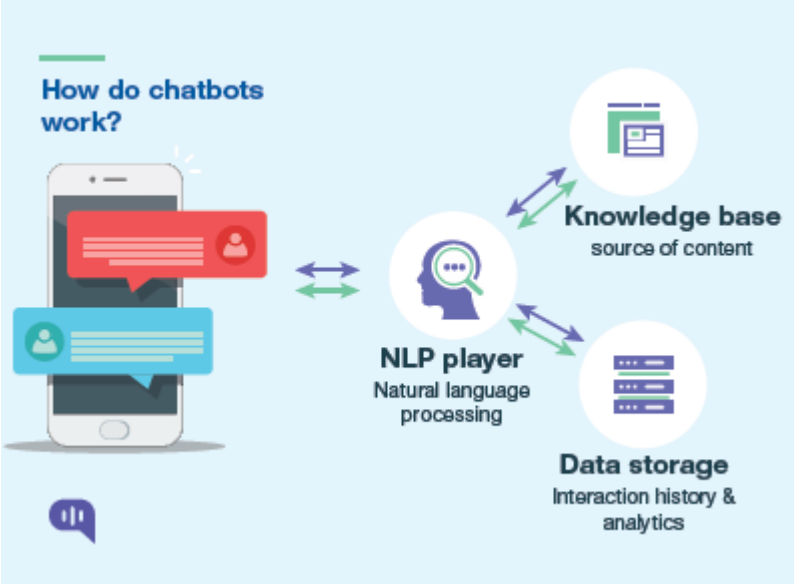


FIGURE 7.8 Chatbot for Booking Appointment

Once the intent of the user's request is correctly known, the chatbot then searches its knowledge base (refer **Fig. 7.9**) to give the most appropriate response. Many-a-time, the user asks for some information or data. In such cases, the chatbot interacts with the back-end system to retrieve the data required to satisfy the user's request.

If there is no information available in the current context or if the chat-bot fails to interpret the user's request, then the chatbot may reply with a generic and predefined text.



Credit: Astrovector / Shutterstock

FIGURE 7.9 Chatbots converse using their knowledge base

7.3.4 Types of Chatbots

Chatbots apply a combination of different AI techniques including natural language processing, machine learning and semantic understanding. There are different types of chatbots (refer **Fig. 7.10**). Each one serves a different purpose and solves a different issue. Depending on the use case you want to address, some chatbot technologies are more appropriate than others.

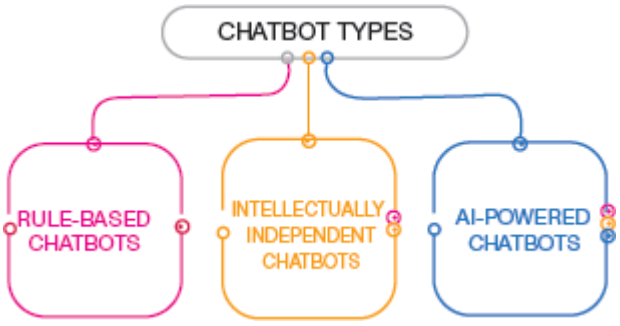


FIGURE 7.10 Types of Chatbot

1. **Rule-based chatbots:** As the name suggests, these chatbots follow predefined paths while conversing with the user. As the conversation progresses, the user is offered several options that correspond to their inquiry (refer **Fig. 7.11**). Based on the option selected by the user, the chatbot decides the next step in the conversation.

   AI-powered chatbots get smarter over time.
   Though these chatbots make the interaction simple for both the user as well as the chatbot program, it can be used only to answer basic customer inquiries (like the status of one's delivery, get some information, etc.).
2. **Menu-based chatbots:** These chatbots are a variant of rule-based chatbots that present a menu or buttons showing different options available to the user. Instead of typing text, the user has to just click on those buttons or options to interact with the system. Such chatbots lack critical understanding and flexibility. These bots can only automate situations that they have learned and are likely to arise.

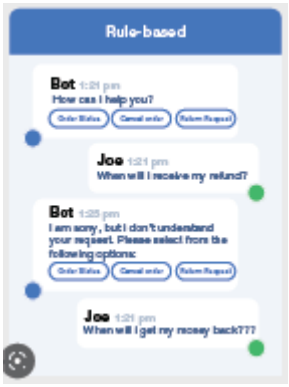


FIGURE 7.11 Rule Based Chatbot

3. **Intellectually-independent chatbots:** These chatbots use machine learning (ML) algorithms to learn from user's requests and bot's reply. During every conversation, the bots use keywords to recognize recurring patterns and common user behaviours to decide what to do when little or no human assistance is available. For example, if the user types 'Facing problems in account login', then the bot breaks the sentence into keywords like 'problem,' 'login,' and 'account,' and then provides a predefined answer based on those keywords.
4. **AI-powered chatbots**: These are hybrid bots that are a good combination of both the bots—rule-based and intellectually-independent chatbots (refer **Fig. 7.12**). These chatbots extensively use AI techniques to

simulate human intelligence, thereby making machines work and think like people. They process user's queries (using neural network) before responding to them. AI-based chatbots also learn from previous experience and use reinforced learning to keep evolving.

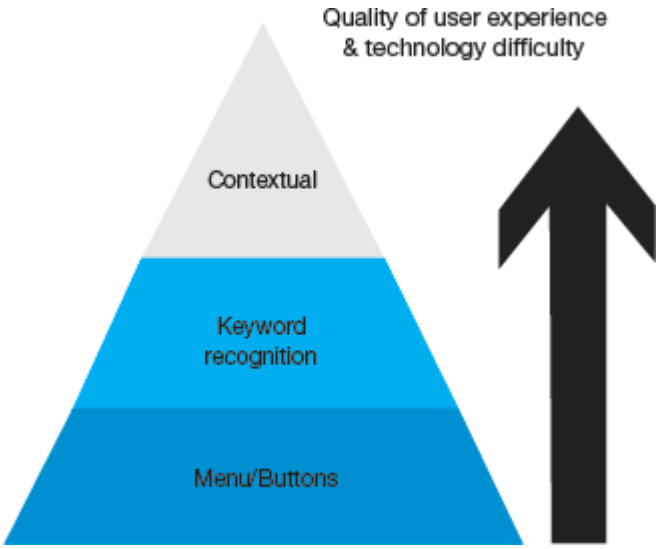


FIGURE 7.12 AI-powered chatbots

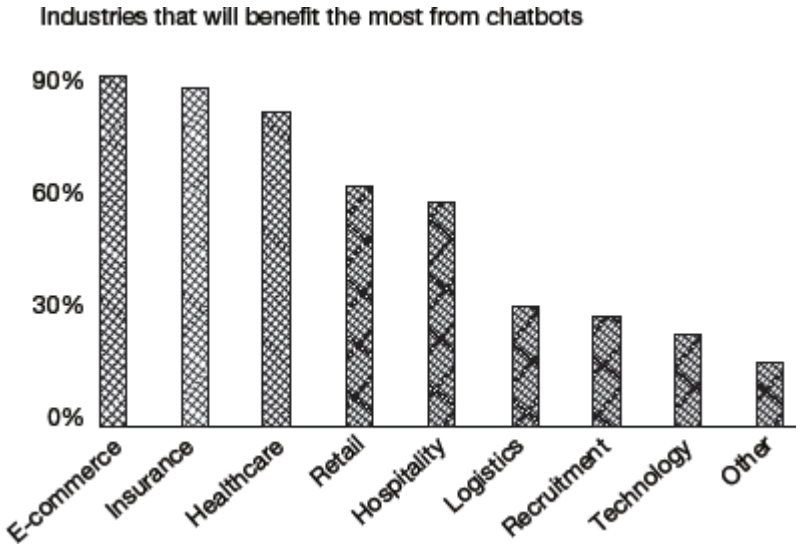


FIGURE 7.13

AI-powered chatbots use a predefined flow to ensure accurate responses to a user's inquiry. If no such path is available, they are also equipped to understand free language. For example, Shivam may give a voice command to a chatbot saying, 'Book me a flight to Chennai.' But Shubham may intend to do the same task by saying, 'I need a flight to Chennai.' Such complexities can only be handled by an AI-based or NLP-based bot

as it identifies the language, context and intent to react accordingly. A rule-based bot only understands a pre-defined set of options.

We can also classify chatbots based on where they are deployed. For example, chatbots can be implemented into messengers, standalone applications or on a website. **Figure 7.13** shows the application of chatbots in different sectors.

7.4 How Does Natural Language Processing Work?

NLP facilitates computers to understand natural language just like humans. An NLP system takes a language as input either in spoken or written form. It processes the language using AI techniques to make sense of it in a way a computer can understand.

Did you know that Facebook's translations achieved superhuman performance in 2019?

Humans have different sensors like ears to hear and eyes to see. Similarly, computers can read content through programs and listen to audio through microphones. While humans process inputs in their brain, computers process in the CPU by following program instructions. During processing, the inputs are converted to codes that can be interpreted by a computer.

7.5 Components of NLP

NLP is a combination of NLU and NLG techniques.

Natural language understanding (NLU) deals with mapping the given input in natural language into useful representations.

Natural language generation (NLG), on the other hand, uses content in internal representation to produce meaningful sentences in a natural language. NLG is performed using the following:

1. ***Text planning*** retrieves relevant content from knowledge base.

2. ***Sentence planning*** selects required words to form meaningful sentences in the right tone in current context.
3. ***Text realization*** finally converts the sentence plan into sentence structure.

NLU is harder than NLG.

TABLE 7.2 Differences between NLU and NLG

NLU	NLG
Reads and interprets language.	Writes or generates language.
Generates non-linguistic outputs from natural language inputs.	Generates natural language outputs from non-linguistic inputs.

7.6 Steps in NLP

From the above discussion, we can summarize the NLP process (refer **Fig. 7.14**) in five steps.

Step 1. Perform Lexical Analysis: Lexicon of a language refers to the collection of words and phrases in that language. Therefore, in this step, the structure of words is identified and analysed by dividing the entire text into paragraphs, sentences and words.

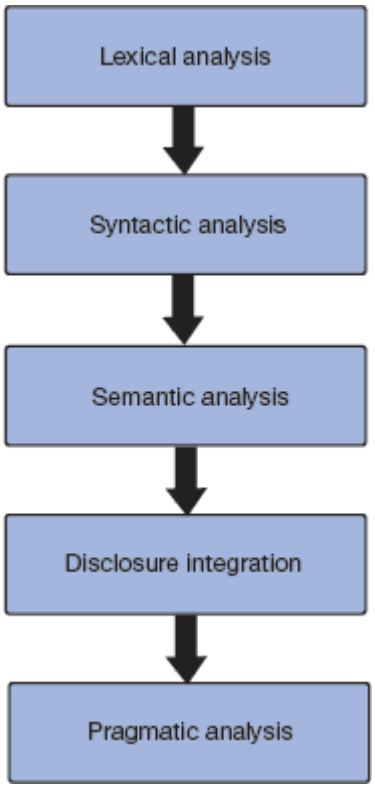


FIGURE 7.14 Steps in NLP

Step 2. Syntactic Analysis or Parsing: As discussed, all the words in a sentence are analysed as per grammatical rules of that language. Words are rearranged to study any relationship that exists between them to ensure that the sentence is syntactically correct. For example, 'The boy was caught by the ball' is rejected by English syntactic analyser.

Step 3. Semantic Analysis: In this step, the exact meaning or the dictionary meaning of a text is drawn. It is done by mapping syntactic structures and objects in the task domain. For example, the phrase 'cold fire' is rejected by the semantic analyser of English language.

Step 4. Discourse Integration: The meaning of a sentence is often studied with respect to the meaning of the sentence just before it. Also, a sentence affects the meaning of sentence immediately succeeding it. Discourse integration studies the meaning of a sentence with respect to meaning of sentence preceding as well as succeeding it in sequence.

Step 5. Pragmatic Analysis: In this step, what was said is re-interpreted on what it actually meant. This is done only in situations that require real world knowledge.

Phonology is the study of organizing sound systematically.

NLP, AI, Machine Learning: What's the Difference?

NLP, AI and ML are important terms that are often used interchangeably. But you must know that, NLP and ML are both subsets of AI.

With NLP, machines can understand what the user wrote or spoke and then respond accordingly. NLP uses ML algorithms to automate the process of delivering accurate responses.

For example, AI-powered chatbots use NLP to interpret what users say and what they intend to do, and ML to automatically deliver more accurate responses by learning from past interactions.

The untapped potential in unstructured text

People's thoughts, research, opinions, facts and feedback transfer into the digital world through social media feeds, legal case files, electronic health records, contact centre logs, warranty claims and more. Natural language processing uncovers the insights hidden in the word streams.

7.7 Phases of NLP

There are two phases in any NLP project—data pre-processing and data processing. In this section, we will learn about both of them.

Data Pre-Processing

In this phase, textual data is prepared and cleansed so that it can be efficiently and correctly analysed by algorithms. Given below are certain ways in which data is prepared for analysis.

1. **Tokenization:** This process breaks down text into smaller units called tokens. There are two types of tokenization.

Sentence tokenization, also known as **sentence segmentation,** divides a **text** written in human language **into** a sequence of individual sentences.
Word tokenization or **word segmentation divides a text** of written language **into** its constituent wor**ds**.

2. **Stop word removal:** Here, all the commonly used words like ‘is, ‘the’, ‘and’, ‘are’, ‘to’, ‘on’, ‘in’, etc. are removed from the text so that unique words that provide most of the information about the text are only studied.

   Morphological segmentation divides words into smaller parts called morphemes. For example, the word untestably is broken into ‘un,’ ‘test,’ ‘able’ and ‘ly’ as morphemes.
3. **Lemmatization and stemming:** Text lemmatization and stemming are used for grammatical reasons to support documents that contain **different forms of a word. For example, words like run, runs, running** are related words having a similar meaning. In this case, **stemming** *and* **lemmatization** *can be used to* **reduce all these related words to** *a* **common base.**
   For example, am, are and is have a common base: ‘be’. Similarly, run, runs, running can be reduced to the base form: ‘run’.
   Stemming and lemmatization are special cases of **normalization**. However, they are different from each other.
   Lemmatization *performs its task with the help of a* **vocabulary** *and* **morphological analysis** *of words, to return the base or dictionary form of a word, which is often known as the* **lemma**.
   S**temmer, on the other hand,** operates **without knowledge of the context. It** cannot understand the difference between words having a different meaning in a different context. However, stemmers are **easier to implement** and usually **run faster**.
4. **POS Tagging** or parts of speech tagging, also known as grammatical tagging, is a process that marks words in text format for a particular part of a speech, depending on its definition and context. It reads text in a language and assigns some specific token (parts of speech) to each word. For example, a word is tagged as noun, verb, adjective, determiners, etc. Like in the sentence, ‘The bird flew.’ bird = noun, flew = verb.

Data Processing

Once the data is pre-processed, it is processed using a particular algorithm. Though there are many NLP algorithms, two main types are commonly used:

1. **Rules-based system:** A rule-based system uses carefully designed linguistic rules for NLP.
2. **Machine learning-based system:** These systems use statistical methods. The ML system is first trained using a training data set. During the learning phase, the system learns how to perform tasks and adjust certain parameters either to achieve higher accuracy or to process more amount of data. These days, usually, a combination of machine learning, deep learning and neural networks is used for NLP algorithms so that the system can automatically hone their own rules through repeated processing and learning.

7.8 Syntax vs Semantic Analysis

Syntax is the arrangement of words in a sentence to make grammatical sense. Syntax of a statement is assessed by checking its meaning from a language based on grammatical rules. During syntactic analysis, processes like tokenization, stemming, lemmatization, stop words removal, POS tagging are performed.

Semantics involves the use of and meaning behind words. NLP applies algorithms to understand the meaning and structure of sentences. Semantics techniques include the following:

1. **Word sense disambiguation:** In this step, the meaning of a word is analysed based on the context in which it is used. For example, consider the sentence, ‘I want to present a watch as her birthday present’. Here, the word ‘present’ has different meanings. An algorithm should be used here to understand that the first ‘present means ‘to give’ and the second ‘present’ means ‘a gift’.
2. **Named entity recognition:** Named entity recognition or NER identifies words that can be classified in different groups. That is, semantics of the text is analysed to differentiate between entities that are visual-

ly the same but otherwise different. NER is especially useful when we have to analyse a piece of text and identify all mentions of a certain company or product. For example, in the sentence, 'McDonald's son, John, went to McDonald's on his friend's birthday.' Here, the algorithm must distinguish between the two instances of 'McDonald's', the first instance is a person and the second instance refers to a restaurant.

3. **Natural language generation:** NLG uses a database to determine semantics behind words and generate new text. For example, an NLP algorithm can automatically summarize a business meeting or an article published about the business. Another example could be to automatically generate news articles or tweets after training the algorithm using certain text.

7.9 Applications of Natural Language Processing

Some applications of NLP are discussed here.

1. **Text classification:** In text classification, tags are assigned to text for categorization in different groups. This is very important for applications like sentiment analysis, which help an NLP algorithm to determine the sentiment, or emotion behind saying/ writing a piece of text. For example, reviews can be analysed using an NLP algorithm to determine if a greater number of users are feeling positive about a new product or writing negative about it. This analysis can be very useful for detecting the intent of potential customers that can, in turn, be used to predict what the speaker or writer may do—buy the product or not, based on the text they are producing.
2. **Text extraction:** Text extraction helps to automatically summarize text and search important pieces of data. For example, keyword extraction retrieves the most relevant words from the text, which can be useful in applications such as the following:
    1. Search engine optimization
    2. Pull out the most frequently used words in the text
    3. Named entity recognition, which extracts the names of people, places and other entities from text.

Google's Director of Engineering Ray Kurzweil predicts that AIs will 'achieve human levels of intelligence' by 2029.

3. **Machine translation:** The NLP algorithm translates text from one language to another without human intervention. For example, we often use Google Translator to convert text from English to Hindi or vice versa.
   Translation tools also facilitate businesses to communicate with customers in their language of choice. This helps in breaking barriers for globalization. Moreover, these translation tools are trained to understand specific terminology that are relevant for a specific industry, like finance or medicine.
4. **Natural language generation:** In this process, algorithms are used to analyse unstructured data and automatically generate content based on that data. For example, GPT3 model analyses unstructured text to generate articles.
5. **Spam detection:** Spam detection technologies use NLP's text classification capabilities to scan emails to check for words that indicates whether it is a spam email or may lead to phishing attack. For example, easy loan, free gifts, bad grammar, threatening language, urgency, misspelled company names, etc. are common points that indicate a spam email.
6. **Spelling correction:** Applications like MS Word support features for spelling correction. This has been possible just because of NLP.
7. **Search engines** use NLP to display relevant search results based on keywords provided by users. This helps even a person who is not very technology-savvy to easily find information that he/she is looking for.
8. **Chatbot:** We all have used chatbots on some website or the other. These chatbots are small AI systems that interact with humans through text or speech. They use NLP algorithms to provide the customer's chat services 24 X 7. This not only speeds up response time but also helps to handle multiple queries simultaneously, thereby freeing up humans from answering repetitive questions.
   Chatbots continuously learn from each interaction. This helps to improve its performance by better understanding user's intents. In case a chatbot is not able to interpret or respond to a user's query, it passes that query to a human.
   For example, chatbots in healthcare, take data from patients to assess their symptoms, and determine next steps to be taken (like set up appointments with the right doctor or recommend treatments).

Technically, chatbots provide automated answers to common customer queries. They use pattern recognition systems with heuristic responses to interact with users. These days, chatbots have evolved from simple answering programs to AI-powered systems that can handle more complicated requests, thereby making conversational experiences increasingly intuitive.

9. *Question-Answer Systems: Question-answer systems are intelligent systems that are specifically designed to respond to customer queries. It is different from chatbots as it has a large knowledge base and possess a good understanding of language. Unlike chatbots, QA systems are far beyond canned answers. They can deal with textual data, audio, video and images. For example, they can answer questions like 'When did India get independent?' or 'How do I get to the airport?'*
   Question-answer systems are present in social media chats and tools such as Siri and IBM's Watson. In 2011, IBM's Watson computer competed on Jeopardy. Jeopardy is a game show during which the contestants ask questions and answers are expected. The entire IT industry was astonished to observe that when computer competed against the show's two biggest all-time champions, it won the first place.

The functions given above are widely used across a range of real-world applications such as the following:

1. Customer feedback analysis to analyse user reviews on social media.
2. Customer service automation uses voice assistants on the other end of a customer service phone line. These assistants use speech recognition to understand customer's query and then to re-direct the call to the appropriate department.
3. Automatic translation like Google Translate, Bing Translator and Translate Me translate text from one language to another.
4. Academic research and analysis analyses a large number of academic material and research papers to create a relevant summary.
5. Analyses and categorizes medical records to predict diseases
6. Word processors use NLP to check plagiarism and do proofreading.
7. NLP algorithms are used to analyse market history and around 10,000 documents to gain an insight into financial trading and predict value of stocks.

8. HR teams these days use NLP to summarize the CVs of applicants and recruit the best talent.
9. AI attorney uses NLP to automate routine litigation tasks.
10. In healthcare, NLP is used to convert sign language into text. This helps deaf people to communicate with people who don't understand sign language.
11. In the aircraft maintenance industry, NLP is used to find relevant information from aircraft manuals (that me 100s of pages) based on problem description given by the pilot.

7.10 Pros and Cons of Natural Language Processing

The most prominent advantage of NLP is that it facilitates efficient interaction between humans and computers. Other benefits include the following:

1. Improved accuracy
2. Enhanced efficiency of documentation
3. Automatic summarization of a large number of unstructured and complex text
4. Interpretation of spoken words by personal assistants like Alexa
5. Chatbots for customer support
6. Perform sentiment analysis
7. Once implemented, NLP solutions are less expensive
8. Save time and effort of human counterparts
9. Faster customer service response times 24 X 7, results in improved customer satisfaction
10. Pre-trained NLP solutions are easily available to facilitate different applications of NLP.

Limitations of NLP Systems

Some limitations of NLP systems are as follows:

1. Training a machine learning model is often time-consuming. Developing and training a new model to get the best results takes several weeks.

2. Remember that no machine learning model is 100% accurate in every situation. Hence, NLP solution may not be reliable. There is always a possibility of errors in predictions.

Challenges

Human languages are often ambiguous for machines as one sentence can have different meanings and several sentences may mean the same thing. Hence, we can say that this difficulty mainly persists due to different levels of ambiguity.

Lexical ambiguity exists due to confusion in different words. For example, whether the word ‘board’ should be treated as noun or as a verb.

Syntax level ambiguity occurs because a sentence can be parsed in different ways. For example, the sentence ‘He lifted the beetle with red cap.’ can be interpreted in two ways. First, did he use cap to lift the beetle? Second, did he lift a beetle that had red cap?

Referential ambiguity is common when using pronouns. For example, Ria met Gauri. She said, ‘I am going to the school.’ Now exactly who is going to school?

NLP is an ever-evolving field. As of now, it is somewhat ambiguous and suffers from certain limitations as follows.

1. **Lacks precision** Human languages are often ambiguous as humans speak using slang and regional dialect. Moreover, their languages are not precise and may be affected owing to social context. On the contrary, computer languages are very precise and highly structured. This variation makes text interpretation a very challenging task indeed.
2. **Tone of voice and inflection** Semantic analysis is not that easy, especially in three cases. First, when humans speak with sarcasm; second, when the meaning of a sentence changes depending on the word or syllable the speaker puts stress on; third, when the speaker changes his/her words while speaking. The tone and inflection of speech is also different when speaking in different accents.

3. **Evolving use of language** Natural languages or human languages are continually changing. Although syntax of the language remains the same, they are subject to change over time.

7.11 Evolution of Natural Language Processing

NLP is a multi-disciplinary field that draws concepts from a number of disciplines including computer science and computational linguistics. Extensive work in NLP is being done from the mid-20th century and its evolution can be understood by analysing major milestones given below.

Augmented Transition Networks (ATN) is a finite state machine that is capable of recognizing regular languages.

1. **1950s** Alan Turing developed the Turing Test to determine whether a computer is truly intelligent or not. In this test, automated interpretation and generation of natural language is considered to be the criteria of intelligence.
2. **1950s–1990s** NLP algorithms were usually rules-based. These rules were developed by linguists to guide how computers would process language.
3. **1990s** The top-down approach to NLP shifted to a more statistical approach with computers becoming faster. Now, the new approach stressed on developing rules based on linguistic statistics without the help of a linguist to create all the rules. Data-driven NLP was the highlight of this decade. However, slowly, instead of just catering linguistics, NLP algorithms shifted from a linguistic approach to an engineer-based approach, that could attract a rich variety of scientific disciplines.
4. **2000–2020s** The field of NLP witnessed a dramatic growth with advances in computing power. The number of real-world applications making use of NLP increased manifold. Now, NLP was extensively used in both the business and consumer spheres, including chatbots, cybersecurity, search engines and big data analytics.

7.12 Handling Ambiguities

There are two main components of NLP—NLU and NLG.

1. **Natural language understanding (NLU):** As the name implies, NLU is done by analysing different aspects of the language. This information, in turn, helps the algorithm to map the input given in natural language into useful representations.
2. **Natural language generation (NLG):** As very much evident from the name, in this component, meaningful phrases and sentences in natural language are produced from some internal representation.
   No doubt, NLU is more difficult than NLG. This is mainly because of two reasons:
   First, natural language has an extremely rich form and structure.
   Second, natural language is very ambiguous.
   Different levels of ambiguity are encountered such as the following:
   1. **Lexical ambiguity** It is the most primitive level of ambiguity and exists at the level of individual words. Lexical ambiguity arises when one or more words in a sentence have multiple meanings.
      For example, is the word “board” a noun or a verb?
      Moreover, consider the sentence- “I saw a bat.” Here, bat can be either the wooden material used to hit ball in cricket or a mammal.
      Similarly, ‘saw’ can be past tense of “see” or the tool used to cut.
   2. **Syntax level ambiguity** It occurs when a sentence can be parsed in different ways. For example, if we say, “He threw the beetle with red cap.” Here, it is difficult to say whether he used a red cap to throw away the beetle or did he throw a beetle that had a red cap?
      Syntactic ambiguity is most commonly occurring ambiguity in English language.
   3. **Semantic ambiguity** Even if meaning of each word in a sentence is clear, there may still be multiple ways in which a sentence can be interpreted. For example, consider the sentences-
      “The dog is chasing the cat.” And “The dog has been domesticated for 10,000 years.”
      In the first sentence, “The dog” means to a particular dog, whereas in the second sentence, it talks about the species “dog”.
      Take another example.

"Nita and Mukesh are married."

This sentence can be interpreted in two ways.

First, Nita and Mukesh are married to each other.

Second, Nita and Mukesh, both are married but not to each other.

4. **Referential ambiguity or Anaphoric ambiguity** Referential ambiguity usually happens when pronouns are used. For example, if we say, 'Goransh called Aarish. He said, 'I am coming to India next week.' Now, who is coming? It is difficult to answer.

   Features like autocomplete are trained to predict the next words in a text, based on what has already been typed.

   Thus, we can say that, a phrase or word refers to something previously mentioned, but there is more than one possibility while deciding what exactly it is. Moreover, sometimes certain things are assumed. For example, in the sentence given below, 'they' actually meant 'hospital', but we know it is actually 'hospital staff' and not 'hospital'.

   "Neon went to the hospital, and they told him that he needs to take ample rest." (They = the hospital staff.)

   Thus, referential integrity exists when, one sentence can have different meanings and when many sentences actually mean the same thing.

5. **Ellipsis:** This occurs when we omit words that are actually needed for grammatical completion of the syntax. The omitted words are easily understood by the reader/listener. Though this is very common in speech, less in writing. For example, 'I am going to Rameshwaram. Also to Kanya Kumari.' This sentence can be understood as 'I am going to Rameshwaram and Kanya Kumari'.

   However, both NLU and NLG need to perform the following:

   Text planning to retrieve only relevant content from the knowledge base.

   Sentence planning to select words to form meaningful phrases while setting the right tone of the sentence.

   Text realization maps the sentence plan into a sentence structure.

7.12.1 Techniques of Ambiguity Resolution

Syntactic constraints can be resolved using the following approaches.

1. *Frequency This is more important especially for resolving lexical ambiguity. In this approach, the most common meaning of a word is preferred. For example, in the sentence, 'I saw a bird.' Here, 'saw' is taken as past tense of see as it is more common than the other meaning 'to cut with a tool'.*
   We can also deduce the meaning of an ambiguous word by considering its frequency in the current context. For example, when we are talking about dinner, the word 'pitcher' means 'container of liquid' but when we are talking about baseball, then 'pitcher' means 'thrower of ball.'
2. *Using Associational frequency. That is, by determining how often two words, or particular meanings of word, appear in text. For example, in the sentence, 'The cheetah and the deer were running very fast. They both could not be seen within a fraction of a second. However, finally it was caught.' In this sentence, 'it' is more likely to be the 'cheetah' because the word 'fast' is usually associated with 'cheetah'.*
3. *Selection restrictions or semantic constraints: This can be applied if we know that concept A can only combine with concept B in mode Z if A or B have specified features. For example, in the sentence, 'The bat ate the flying insects.' We can easily deduce that 'bat' refers to a mammal here because only living objects can eat. So, this 'bat' is not the wooden material to play cricket.*
   Similarly, the sentence, 'The broken bat is kept there.' Here, the adjective 'broken' usually refers to non-living things. So here, it is the wooden material to play.
4. *Recency rule: This rule is usually used for resolving anaphora. In this approach, it is believed that in 85% of the cases, the referent is the most recently mentioned object of correct gender and number.*
5. *Parallel structure. Again, this technique is used to resolve issues related to anaphora. For example, 'Neha went with Muskan, and she drove her back.' In this sentence, she is 'Neha' and her is 'Muskan'.*
6. ***World knowledge*** In this technique, the most plausible interpretation is used. For example, 'Cherry got down for lunch, and so did Mary.'

7. *Textual coherence: Here, the most plausible meaning is considered. For example, in the sentence,*
   'Myra said that people should vote for Kiran in the upcoming elections. She said that she is the best amongst all other candidates.'
8. *Speaker intent. In this approach, the text is interpreted in terms of speaker's intent. For example, 'You were to be here by 9, do you know what time it is?'. Another example, "Excuse Me! What did you say?" Note that in both the sentences, the speaker is not actually asking. The intent is to make the other person realize something. So, in computing perspective. these ambiguities can be resolved by understanding the context in which they were spoken.*

7.13 The NLP Model of Perception

According to the Oxford dictionary, perception is the ability to see, hear or become aware of something through senses. Humans perceive language not only from text, but from all our senses. For example, we visualize a situation just by listening to the tone of the speaker; we listen to the tone of the voice, we visualize the situation. Similarly, we use senses like touch and smell and even non-verbal communication like body language to interpret language.

Consider the adjective 'blue' that has various meanings. For example, 'blue sea' means the colour of the sea is blue. When we say, 'he is feeling blue', it means that the person is feeling sad. Similarly, when we say that a person's face turned red, then that means the person was angry or ashamed and does not mean literally red colour. To correctly interpret these words, we need an NLP model that can identify and deal with such complexities.

The current NLP models can understand the context from the text only. This is a serious limitation and to make the NLP models perform at the human level, we need to break these barriers. Bisk et al. 2020 said that perception should be an important dimension in NLP models and it can be derived from the different types of inputs including audio recordings, images and videos. The body language or the voice tone, are currently at

the sideline, of the NLP research. But the creation of these new benchmarks is expected in the next few years within NLP research.

7.13.1 NLP Communication Model

The 'NLP communication model' (refer **Fig. 7.15**) is the basis for how human beings perceive reality. In fact, all advances of neuro-semantics were discovered with this model as the foundation as it provides a basic framework for mapping what happens in a person's mind-body system in between stimulus and response.

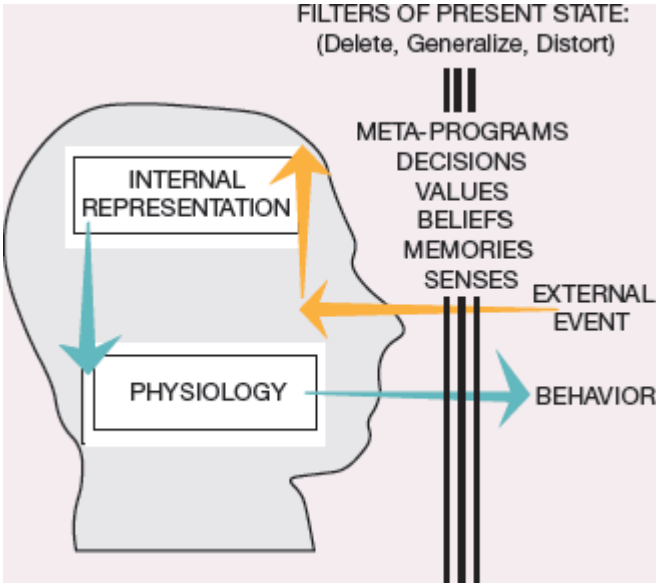


FIGURE 7.15 NLP communication model

From 'External Event' to 'Internal Experience'

The moment we receive a stimulus, our initial *experience* of the 'external world' begins. These external events (or stimuli) are converted into neuro-linguistic (mind-body) 'information'.

The Information is Filtered Through Our State

The information about external world (stimuli) is then filtered our present mind-body state. For example, if someone disturbs you intentionally while doing an important work, then you will respond differently if your work is going well and in a different manner if you are already under stress.

Thus, we think and perceive things differently depending on our state of mind that in turn is affected by the filters including memories, decisions, beliefs, values, identities, thinking patterns, etc.

Internal Representation

After the 'filtering process', we get the internal representation of the event/object in the world which is based on our personal perception of it.

State

The internal representation then feeds down into our physiology and combines with present state to form a new present state. Note that this is all done in response to the internal representation of the external event (stimulus).

Thus, the state of our mind and body **govern our behaviour** and we respond to the world by **filtering down the mental map of the world based on our available internal representations.** Therefore, we need to manage our states well that open us up to new responses to perform at our best and to unleash our highest potential.

Perceptual Positions

Perceptual position is a form of modelling in which one steps into somebody else's shoes, and sees what the other person sees, hears and feels. This helps us to get an impression of the client's reality.

Generally, when we imagine ourselves as someone else, stand/sit in the other person's posture, speak with their voice, our mirror neurons get triggered and we actually pick up a lot of information about what how he/she feels thereby improving our interpersonal results.

7.13.2 The Five Perceptual Positions

We can adopt one of the five perceptual positions to fully understand another's point of view. These positions are as follows:

1st position is where we are ourselves *associated to interpret* things from our own point of view.

2nd position is where we *dissociate* from ourselves and image ourselves in someone else's body to understand things from *their* point of view.

3rd position is where we take the role of an *independent observer* to analyse things from an outside point of view.

4th position is observing the 3rd position in relationship to the 2nd position to understand how neutral we as an observer are.

5th position is where we observe how we relate ourselves to an observer.

Note that 4th and 5th positions occasionally used.

Example: If you have an important meeting coming up with a client, then let us see these five perceptual positions.

1st position is when you imagine a 30 second movie of the meeting from **your** perspective and then describe the meeting in one word.

2nd position is when you imagine yourself **as the client** and then replay the movie to describe the meeting from the client's perspective.

3rd position is when you imagine yourself as **an independent (invisible) observer** and replay the movie to understand the behaviour of the participants and learn how it affects the atmosphere of the meeting. For example, were the participants aggressive, friendly, active/passive or respectful.

In the 4th position, you go back to the 2nd position and play the movie from the **client's perspective** to identify if *anything changed and* what could have been done to improve the meeting. Then you need to incorporate the improvements and then play the movie again in the 1st position to observe what has changed.

A Case Study: Image Analysis and Nlp In Healthcare

In healthcare, a number of medical images are created including X-rays, computational tomography (CT) scans, computed axial tomography (CAT), magnetic resonance imaging (MRI) and modern nuclear imaging modalities such as PET and SPECT.

While image analysis is the process of extracting useful information from digitally stored images using different processing techniques, NLP, on the other hand, applies computational techniques to the analysis and synthesis of natural language and speech. In the healthcare industry, patient records contain a lot of important data like medication, immunization records and lab results. But the problem is that this information is not available as structured text presented in rows and columns and extracting information from it is a difficult task.

NLP techniques are used to process, analyse and interpret clinical text that allow medical professionals to ask important questions regarding an individual's health. NLP can then be used to perform tasks like sentence boundary detection, tokenization, problem-specific segmentations, spelling or grammatical error identification and recovery, word sense disambiguation, information extraction, multi-atlas segmentation, fuzzy clustering, graph cuts, genetic algorithms, support vector machines, random forests and deep learning to improve image analysis. Then, speech-to-speech translation engines, social-media mining about health, sentiment analysis toward products and services are added advantages.

However, some of the challenges in this field are as follows:

- Need for generic image analysis techniques that can be efficiently adapted for a specific clinical task.
- Algorithms for analysing heterogeneous image data.
- Models from medical images with less user interaction.
- To understand the language's context and reasoning just like humans.
- Acquisition of the data for image analysis and NLP has lot of ethical issues. Patients have the right to decide how and when their data is used. However, the more the data is available for research, the more advances can be made in the field of image analysis and NLP.

There is a need to develop a diagnostic system that can efficiently read pathology reports by analysing images and the textual annotations that come on the reports, to generate a textual report. For example, the system can analyse the bone scan image and generate a report on the probability of how well treatment of cancer in case of bone cancer. By analysing the text from other clinical inputs such as lab reports and clinical consultation notes in addition to the bone scan image, the system can recommend further treatment for the patient.

The system can, in fact, analyse images from a large number of patients and stratify the population based on some clinical criteria, to help policy decision makers formulate strategies based on the results obtained by analysing reports.

Case Study: How to Automatically Generate Textual Descriptions for Photographs with Deep Learning

To give a caption to an image, we need to generate a human readable textual description. Though this task is easy for a human, it is not trivial for a machine as it involves application of computer vision and NLP techniques.

To describe an image (refer **Fig. 7.16**), the content of the image needs to be understood and translated to meaning in the terms of words. Let us see the tasks involved in doing this activity.

Classify Image

The computer assigns a class label to an image from one of many known classes (as shown in **Fig. 7.17**).

Describe Image

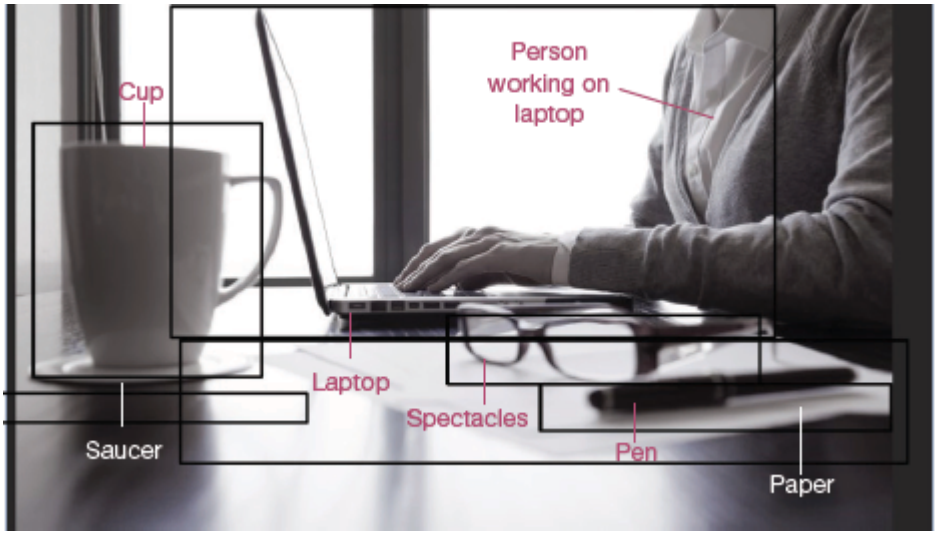
Generate a textual description of the contents image.

Annotate Image

Generate textual descriptions for specific regions on the image.



FIGURE 7.16 Images with Textual Description



Credit: KieferPix / Shutterstock

FIGURE 7.17 Classify Image

Neural Captioning Model

Neural network models are extensively used to automatically generate captions for images. This is done using feature extraction and language model (given in **Fig. 7.18**), that are discussed below.

Feature Extraction Model

The feature extraction model is a neural network-based approach that will extract the salient features from an image, which is usually a fixed-length vector. The resultant vector gives an internal representation of the image which cannot be directly interpreted by the end-users. For image analysis, deep convolutional neural network (CNN) model is used that is already trained on the images stored in the image captioning dataset.

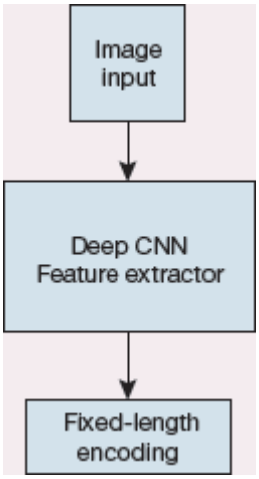


FIGURE 7.18 Feature Extraction Model

Language Model

A language model analyses the words in a sequence to predict the probability of the next word in that sequence. For the task of giving captions to an image, the language model uses the extracted features from the neural network to predict what words should appear in the description of the image. For prediction, we usually use a recurrent neural network, such as a long short-term memory network, or LSTM, as the language model. Each output generates a new word in the sequence which is then encoded and passed to the decoder as input for generating the subsequent word.

An improved form of the model gathers the probability distribution of words across the vocabulary to form the output sequence. The output sequence is then searched to generate multiple possible descriptions. These descriptions are finally rated by likelihood. For quick and accurate results, the language model is trained using pre-computed features extracted from the image dataset.

Note that a separate encoder-decoder is required for translation in a different language. So, the architecture that worked for English will not work for French.

Thus, the entire encoder-decoder architecture model is *based on a CNN that encodes an image into a compact representation, followed by a recurrent neural network (usually* a stack of LSTM layers as shown in **Fig. 7.19**) *that acts as a decoder and generates the corresponding sentence. The model is trained to maximize the likelihood of the sentence given the image.*

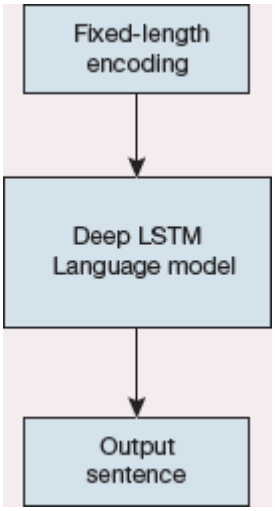


FIGURE 7.19 CNN with LSTM layers

Recent advances in image analysis, NLP and caption generation have made it possible to identify salient parts of an image while generating its caption.

7.14 Constituency Grammar

In constituency grammar, a sentence can have basic constituents as any word or a group of words (also called phrase). These constituents are organized in a sentence based on their properties. These properties can be studied either by using part of speech tagging (PoS) or by using a noun or verb phrase identification.

We know that grammar consists of certain rules. These rules have a single symbol on the left-hand side of the rewrite rules. For example, let us create grammar to parse a sentence –

'The girl sang a song'

Articles (DET) – a | an | the

Noun Phrase (NP) – Article + Noun | Article + Adjective + Noun

= DET N | DET ADJ N

Verb Phrase (VP) – NP V | V NP

Now, we will create a parse tree that will break down the sentence into its structural parts. This will help the machine to easily understand and process the information. To help the parsing algorithm to construct this parse tree, a set of rewrite rules is used to describe what structures are legal in that language.

These rules dictate how a certain symbol in the tree can be expanded by a sequence of other symbols. Given below are some rules.

S → NP VP

NP → DET N | DET ADJ N

VP → V NP

For example, as per the first rule, a sentence (S) can be formed using a noun phrase (NP) and a verb phrase (VP) (given in **Fig. 7.20**).

The second rules say that a noun phrase is a group of words that either begins with a determiner (DET) and is followed by a noun (N) or begins with a determiner (DET), followed by an adjective (ADJ) and then a noun (N).

Similarly, the third rule states that a verb phrase (VP) is formed by combining a verb (V) and a noun phrase (NP).

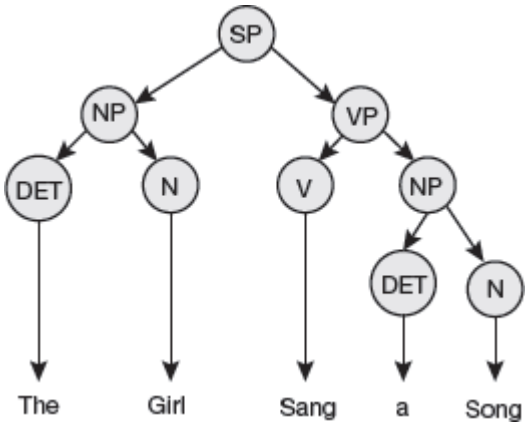


FIGURE 7.20 Sentence with Noun Phrase followed by a Verb Phrase

According to the sentence, lexicon can be formed as follows:

Nouns – girl | girls | song | songs

Verbs – sing | sang | singing

Adjectives (ADJ) – beautiful | patriotic | sad

Using the above grammar and lexicon, we can create the parse tree as given in **Fig. 7.20** . However, considering the above rewrite rules, V can be replaced by both, sing or sang. So, any sentence like 'The bird sing a song' will also be accepted even though it is not correct.

Merits

1. It contains the most popularly used grammar.
2. It is very simple.

Demerits

1. This grammar is not very accurate. As we have seen, the sentence, 'The girl sing a song' is syntactically correct according to parser, but is otherwise incorrect.
2. To enhance accuracy, multiple sets of grammar are required. While one set of rules is used to parse singular nouns/verbs, another set can parse plural forms. Similarly, a completely different set of rules can be used to parse passive sentences, and another ruleset to parse active

sentences, so and so forth. However, this can result in a huge set of rules that after a point in time becomes unmanageable.

Example: Observe the parse tree given in **Fig. 7.21** and notice how the sentence is formed.

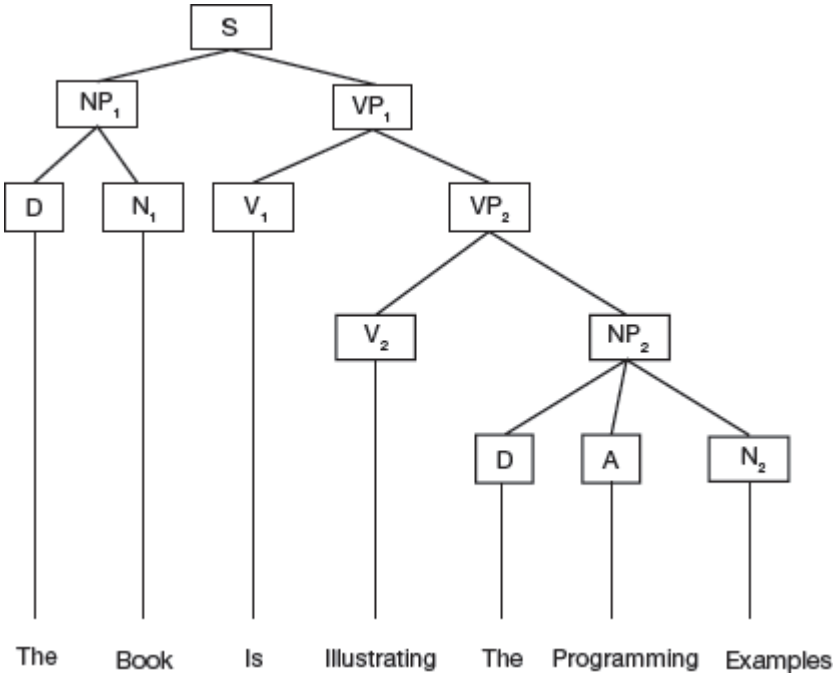


FIGURE 7.21 Parse Tree for the given sentence

7.15 Context-Free Grammar

Context-free grammar is a formal grammar that can be used to generate all possible strings in a given formal language. CFGs were developed by Noam Chomsky in the mid-1950s to help users specify rules that state how a constituent can be segmented into smaller and smaller constituents, up to the level of individual words. Its simple yet powerful formalism could describe the syntactic structure of natural languages.

Mathematically, a context-free grammar G can be defined using a 4-tuple (N, T, S, P) value, where

N or V_N is the set of non-terminal symbols, or variables.

T or ∑ is the set of terminal symbols.

S refers to the Start symbol and naturally, S ∈ N

P is the set of Production rules that dictates how T and V can be combined. Every production rule consists of non-terminals, arrow and terminals.

The left side of production is non-terminals while the right side is called terminals. Such a rule is of the form $\alpha \rightarrow \beta$, where α and β are strings on $N \cup \Sigma$ and at least one symbol of α belongs to V_N.

Technically, CFG is a superset of regular grammar as shown in **Fig. 7.22**.

7.15.1 Derivations

A derivation is a sequence of rule applications that derive a terminal string $w = w_1 \ldots w_n$ from S. This means that, in CFG, the start symbol derives the string by repeatedly replacing a non-terminal by the right-hand side of the production rule, until all non-terminals have been replaced by terminal symbols (refer **Fig. 7.23**).

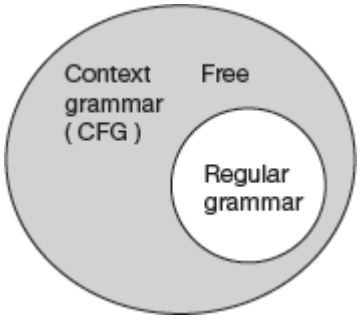


FIGURE 7.22 Relationship between CFG and Regular Grammar

Image Source: Google Images

S	Rule Applied
aS	Rule 1
aaS	Rule 1
aaaS	Rule 1
aaaaS	Rule 1
aaaaaS	Rule 1
aaaaaaS	Rule 1
aaaaaa	Rule 2

FIGURE 7.23 Deriving String from Start Symbol

Example: Given below are some production rules, derive the string "abbcbba" using the CFG.

Rule 1: S → aSa

Rule 2: S → bSb

Rule 3: S → c

Step 1: Let's start with rule 1, S ⊠ aSa

Step 2: Using Rule 2, we can get the string S ⊠ abSba

Step 3: Again, using Rule 2, we can get the string S ⊠ abbSbba

Step 4: Now, using Rule 3, string S ⊠ abbcbba

Example: Construct CFG for a language that permits any number of as.

Solution: We can write regular expression of such a language as, **Regular Expression= a⋆**

Let us now list the production rule for our regular expression.

Rule 1: S -> aS

Rule 2: S -> ε

To derive a string “aaaaaa”, we will start with the start symbol and proceed as given in Table.

Thus, the regular expression, a* is capable of generating a set of strings {ε,a,aa,aaa,...}

Here, we have written, the null string (ε) because S is a start symbol and by applying Rule 2 we can straight forwardly get null string.

7.15.2 Using Nominals in CFG

Nominals contain the head and any pre- and post- modifiers of the head.

Pre-quantifiers include cardinals and ordinals (like four desks), adjectives (e.g., big data) and ordering constraints (like seven small dwarves).

Postmodifiers are of three types:

Prepositional phrases like tea from India.

Non-finite clauses like trains departing after noon.

Relative clauses like restaurants that serve Jain food.

Some general (recursive) rules to handle nominals are,

Nominal → Nominal PP

Nominal → Nominal GerundVP

Nominal → Nominal RelClause

We can even refine noun phrase (NP) as follows:

NP → DET NOM; for example, a flight

NP → Proper Noun; for example, Chennai

NOM → NOUN | NOM NOUN; for example, flight | morning flight

NP → Pronoun; for example, I

Points to remember:

1. PP stands for prepositional phrase and is equivalent to proposition NP. For example, from Hyderabad.
2. VP stands for verb phrase and is equivalent to
   VP → Verb; for example, want.
3. VP → Verb NP; for example, wants a chocolate
4. VP → Verb NP PP; for example, want a chocolate from Switzerland
5. VP → Verb PP; for example, arriving on Monday

Example: Use CFG for parsing the sentence, “I prefer a morning flight”.

Step 1: Let’s start with S

Step 2: Expand S as NP VP

Step 3: Replace NP with Pro. So, now we have, PRO VP

Step 4: Replace Pro with I. We get, I VP

Step 5: Replace VP with Verb NP. Thus, it is now, I V NP

Step 6: Replace prefer with V to get, I prefer NP ***Step 7:*** Replace NP with Det Nom. So, I prefer DET NOM

Step 8: Replace DET with ‘a’. Therefore, we are left with, I prefer a NOM

Step 9: Expand NOM by replacing it with NOM NOUN. It becomes I prefer a NOM NOUN.

Step 10: Replace NOM with NOUN. That make the expression, I prefer a NOUN NOUN

Step 11: Replace morning with first NOUN. We get, I prefer a morning NOUN.

Step 12: Finally, we get, I prefer a morning flight.

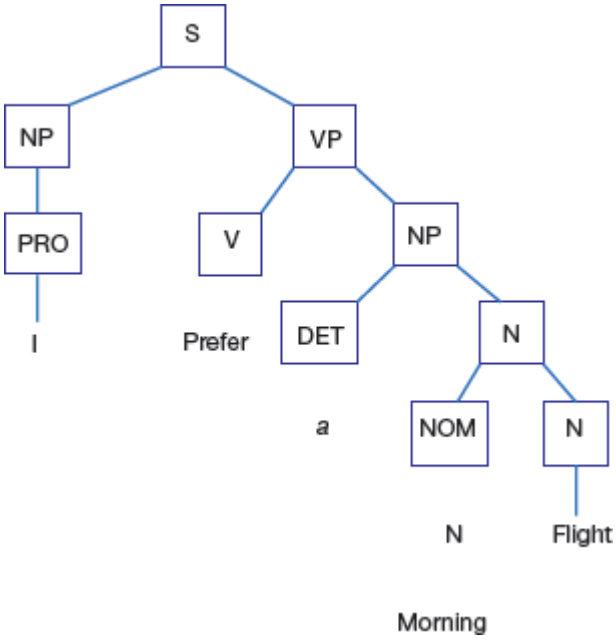


FIGURE 7.24 Parse Tree for “I prefer a morning flight”

Example: Create a parse tree for the sentence, “All the late-evening buses from Delhi to Rhishikesh leave at midnight”

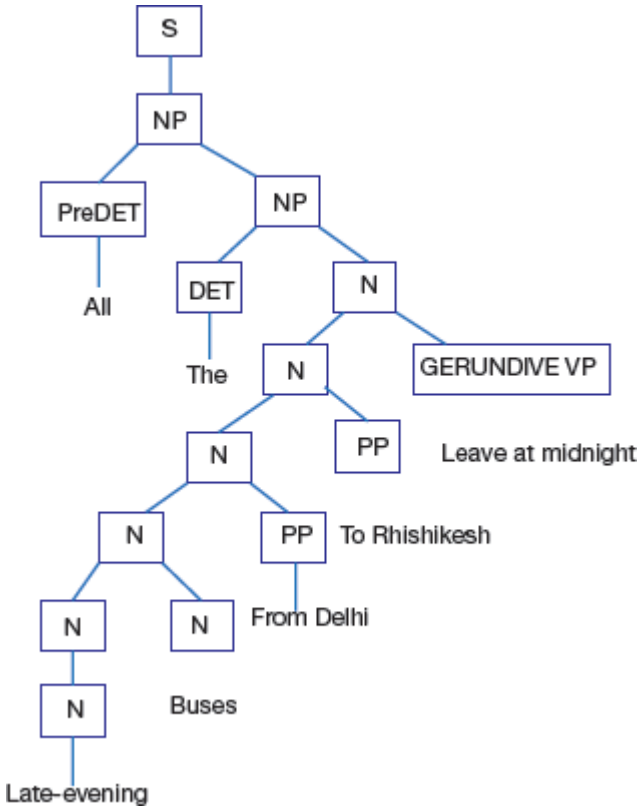


FIGURE 7.25 Parse Tree for "All the late-evening buses from Delhi to Rhishikesh leave at midnight"

7.15.3 Uses of CFG

CFGs can be used in the following scenarios:

1. Describe programming languages.
2. Automatically construct an efficient parser for a properly designed grammar.
3. With the help of information about associativity and precedence, suitable grammars can be constructed for expressions.
4. Describe nested structures like balanced parentheses by matching the starting and closing brackets, checking if-then-elses, etc.

7.15.4 Types of Derivation

We have seen that derivation specifies a sequence of production rules. During parsing, two decisions have to be taken:

First, which non-terminal has to be replaced.

Second, the production rule by which the non-terminal will be replaced.

To decide which non-terminal has to be replaced with production rule, we have two strategies.

1. **Left-most Derivation:** In this derivation, the input string is read starting from the left. At each step, the input scanned is replaced with the chosen production rule.

   Example: Given the input string a + b * c and production rules as given below, demonstrate left-most derivation.

   Rule 1: S = S * S

   Rule 2: S = S + S

   Rule 3: S = a | b |c

   Step 1: Apply Rule 1, to get, S = S * S

   Step 2: Apply Rule 2 on left S after '=', to get, S = S + S * S

   Step 3: Apply Rule 3, to replace leftmost S after '=' with 'a', to get, S = a + S * S

   Step 4: Apply Rule 3, to replace leftmost S after '=' with 'b', to get, S = a + b * S

   Step 5: Apply Rule 3, to replace leftmost S after '=' with 'c', to get, S = a + b * c

2. **Right-most Derivation: T**his type of derivation is just the opposite of left-most derivation. Here, the input is scanned from right to left and replaced with an appropriate production rule.

   Example: Given the input string a + b * c and production rules as given below, demonstrate right-most derivation.

   Rule 1: S = S * S

   Rule 2: S = S + S

   Rule 3: S = a | b |c

   Step 1: Apply Rule 2, to get, S = S + S

   Step 2: Apply Rule 1 on rightmost S after '=', to get, S = S + S * S

   Step 3: Apply Rule 3, to replace rightmost S after '=' with 'c', to get, S = S + S * c

Step 4: Apply Rule 3, to replace rightmost S after '=' with 'b', to get, S = S + b * c

Step 5: Apply Rule 3, to replace rightmost S after '=' with 'a', to get, S = a + b * c

7.15.5 Parse Tree

We have seen that parse tree is a graphical representation of terminal as well as non-terminal symbols. While leaf nodes represent terminal nodes, interior nodes, on the other hand, are the non-terminal nodes. We can derive an entire string by parsing the tree starting from the root node. This root node contains the start symbol.

It is also interesting to note that parse tree follows the precedence of operators. The deepest sub-tree is traversed first. Therefore, an operator in the sub-tree has more precedence than an operator in the parent-tree.

Usually, the tree is parsed using in-order traversal method. In this method, first left-subtree is parsed, followed by the root node which is then followed by traversal of the right-subtree.

Example: Consider the production rules given below to generate the parse tree for the expression a + b - c

Rule 1: S = S + S | S * S

Rule 2: S = a | b | c

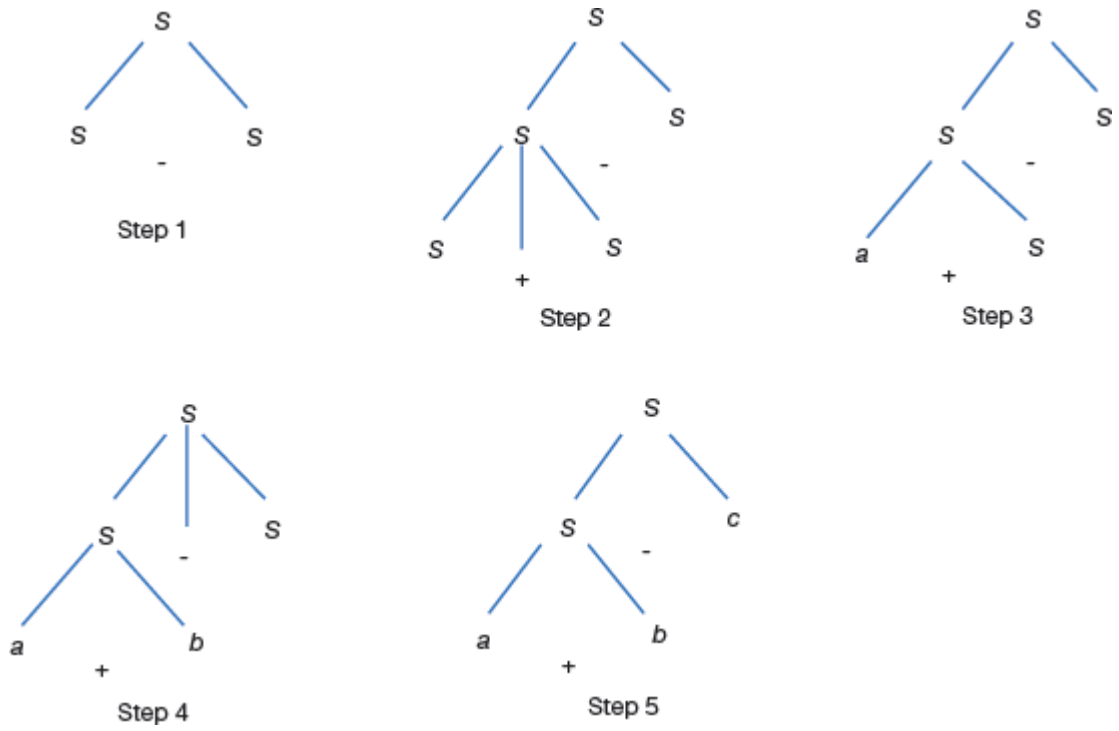


FIGURE 7.26 Parse Tree for expression, a + b - c

7.15.6 Ambiguity in CFG

A grammar is said to be ambiguous if there exists more than one leftmost derivation or more than one rightmost derivative or more than one parse tree for a given input string. Correspondingly, a grammar that is not ambiguous is known as unambiguous.

Example: Given the input string, aabb and production rules given below. Prove that this grammar is ambiguous.

Rule 1: S = aSb | SS

Rule 2: S = ϵ

For the string aabb, we can generate more than one parse tree. Three of them are given below in **Fig. 7.27**.

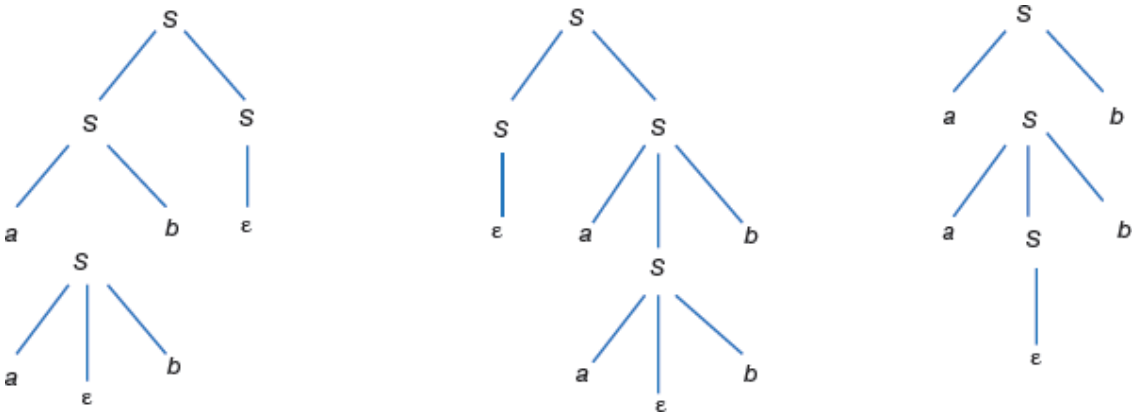


FIGURE 7.27 Parse Tree for the given grammar

Hence, this grammar is ambiguous and thus not good for a compiler construction. To remove this ambiguity, we need to re-write the grammar. There is no automatic tool to detect and remove such ambiguity.

7.15.7 Parser

Parser is a compiler that breaks the data coming from lexical analysis phase into smaller elements.

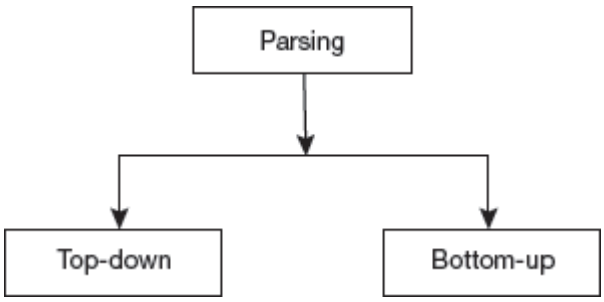


FIGURE 7.28 Techniques for Generating Parse Trees

As input, the parser accepts a sequence of tokens to generate output in the form of a parse tree. A parser can produce the parse tree either by using top-down parsing or bottom-up parsing technique (refer **Fig. 7.28**).

Top-down paring, also known as recursive parsing or predictive parsing, starts the parsing process from the start symbol and transforms it into the input symbol. Bottom-up parsing, on the other hand, constructs the parse tree for an input string. It starts with the input symbol and construct the parse tree up to the start symbol. For this, rightmost derivation is used to

trace the string in reverse. Bottom-up parsing is also known as shift-reduce parsing.

Example: Consider the Parse Tree representation of input string “abcd” using top-down approach.

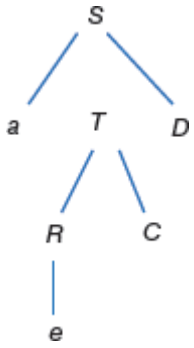


FIGURE 7.29 Parse Tree for “abcd” using top-down approach.

Example: Using the production rules given below, construct a parse tree using bottom-up parsing technique.

Rule 1: E → T

Rule 2: T → T + F

Rule 3: T → a

Rule 4: F → T

Rule 5: F → a

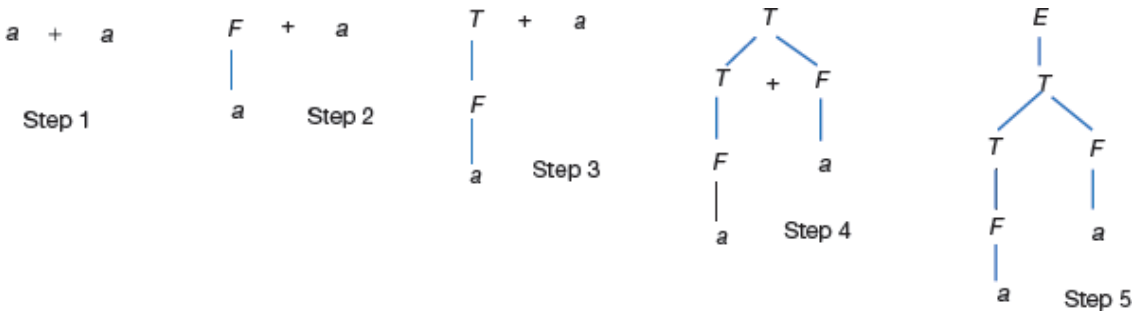


FIGURE 7.30 Parse tree using bottom-up parsing

7.16 Speech Recognition

Speech recognition, or speech-to-text, is the ability of a machine to identify words spoken aloud to convert them into readable text. Speech recognition software usually have a limited vocabulary and may identify words only when they are spoken loud and clear. A sophisticated can process words spoken in different accents and languages.

Speech recognition includes concepts from several disciplines including computer science and linguistics. These days, many modern devices that work on text-focused programs are being equipped with speech recognition functions to facilitate easier or hands-free use of the device.

Speech recognition is often thought of as voice recognition. But they are two different technologies. **Table 7.3** differentiates the two terms.

TABLE 7.3 Differences between Speech recognition and Voice recognition

Speech Recognition	Voice Recognition
• Identifies words in spoken language. • Speech recognition translates anyone's voice. • Accuracy rate is low.	• It is a biometric technology that identifies an individual's voice. • Voice recognition understands a specific user's voice. • Exhibits higher accuracy.

Note that a voice recognition system depends on a recorded template of a user's voice, and is called 'template matching.' The software is trained to recognize a user's voice. To understand how it works, consider the steps given below.

Step 1: The program will show printed word(s) spoken by the user and repeats several times into the system's microphone to train the voice recognition software.

Step 2: Statistical average of multiple samples of the same word(s) are calculated.

Step 3: The average samples as a template in its data structure.

7.16.1 How Does Speech Recognition Work?

Speech recognition systems process spoken words using computer algorithms to accurately interpret them so that they can be converted into text. For example, when a software program converts the sound recorded by a microphone into written text that is readable and understandable by humans, the following these steps are performed.

Step 1: Analyse the audio.

Step 2: Break the audio into smaller parts.

Step 3: Digitize the individual parts into a computer-readable format.

Step 4: Appy a suitable algorithm to match to convert into most accurate text representation.

Speaker-dependent voice recognition devices can provide personalized responses to a user.

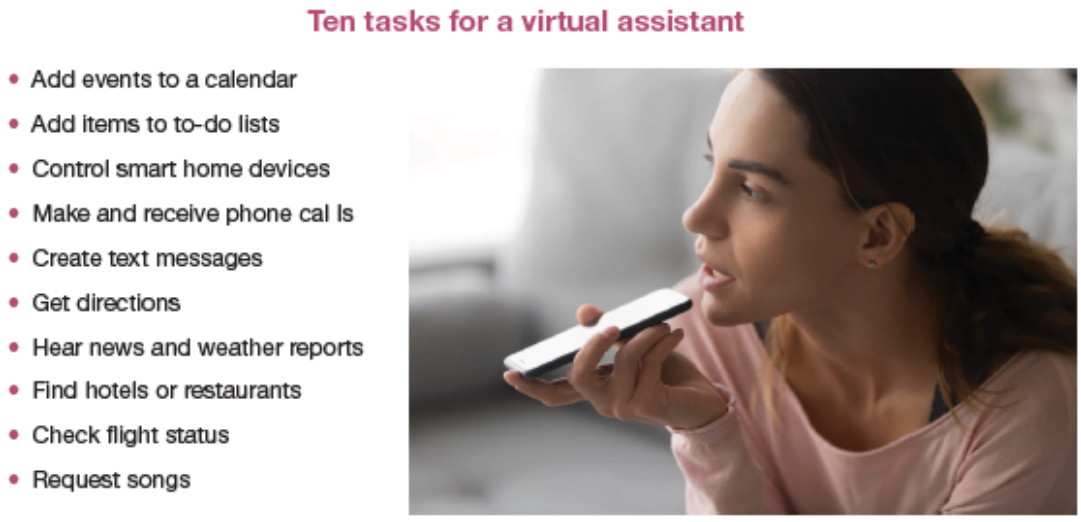
A sophisticated speech recognition software is designed to handle the highly variable and context-specific nature of human speech. The software algorithms are trained on different speech patterns, speaking styles, languages, dialects, accents and phrasings so that it can accurately convert audio into text. The software also delineates background noise that often accompanies the audio. To meet these requirements, a speech recognition software is developed using two types of models:

1. Acoustic models to represent the relationship between linguistic units of speech and audio signals.
2. Language models to match sounds with word sequences to differentiate between words that sound similar.

7.16.2 Applications of Speech Recognition

Some of the popular applications of speech recognition systems include the following.

1. **Mobile devices.** These devices use voice commands for call routing, speech-to-text conversion, voice dialing and voice search. Now, users can respond to a text message without even looking at the devices. These days, speech recognition has become so popular that even PDF document readers and word processing applications like Microsoft Word have speech recognition as an in-built capability to facilitate users to dictate words that can be converted into text and vice versa.
2. **Virtual assistants** use speech recognition to communicate with users and perform different tasks by giving voice commands (as shown in **figure 7.31**).
3. Users can use speech recognition software to learn the correct pronunciation of a given set of words in a particular language.

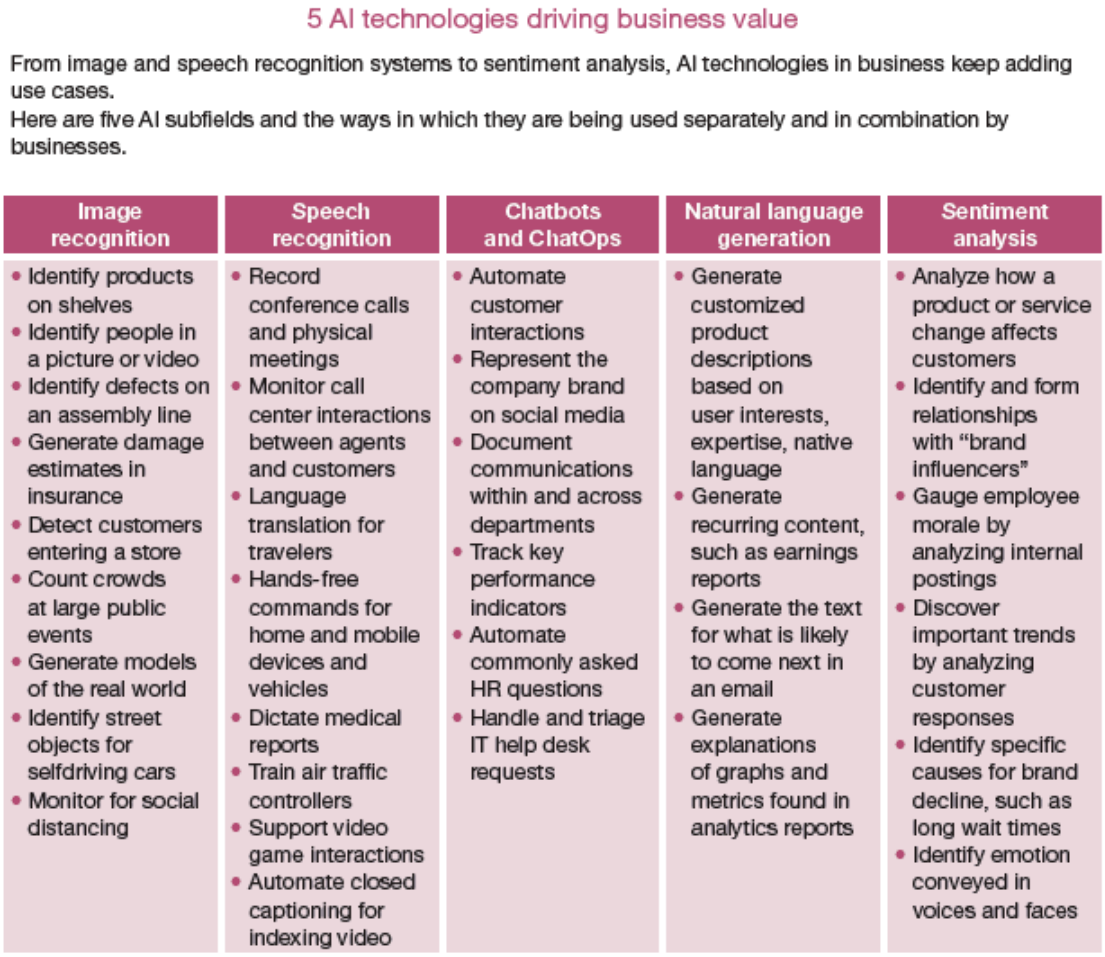


Credit: fizkes. Shutterstock

FIGURE 7.31 Ten tasks for a virtual assistant

4. Speech recognition software is used to provide 24 X 7 customer service using automated voice assistants that listen to the customer's queries to provide relevant information.
5. Doctors use real-time speech recognition software to transcribe notes that can be stored in the healthcare records. Similarly, courtroom reporting is done by such software rather than by human transcribers.
6. Speech recognition software is used by people with hearing disability to translate spoken words into text so that they can easily understand what others are saying. The software is also a boon for those who can use their hands in a very limited way to use computers by giving voice commands.
7. Speech recognition, when paired with sentiment analysis, can be used to identify the emotion(s) of the speaker about a product or service.

8. Speech recognition software is used for hands-free communication by drivers who can control phones, radios and global positioning systems without even touching them. This gives drivers a better control on their vehicle thereby reducing the risk of accidents.



5 AI technologies driving business value

From image and speech recognition systems to sentiment analysis, AI technologies in business keep adding use cases.
Here are five AI subfields and the ways in which they are being used separately and in combination by businesses.

Image recognition	Speech recognition	Chatbots and ChatOps	Natural language generation	Sentiment analysis
• Identify products on shelves • Identify people in a picture or video • Identify defects on an assembly line • Generate damage estimates in insurance • Detect customers entering a store • Count crowds at large public events • Generate models of the real world • Identify street objects for selfdriving cars • Monitor for social distancing	• Record conference calls and physical meetings • Monitor call center interactions between agents and customers • Language translation for travelers • Hands-free commands for home and mobile devices and vehicles • Dictate medical reports • Train air traffic controllers • Support video game interactions • Automate closed captioning for indexing video	• Automate customer interactions • Represent the company brand on social media • Document communications within and across departments • Track key performance indicators • Automate commonly asked HR questions • Handle and triage IT help desk requests	• Generate customized product descriptions based on user interests, expertise, native language • Generate recurring content, such as earnings reports • Generate the text for what is likely to come next in an email • Generate explanations of graphs and metrics found in analytics reports	• Analyze how a product or service change affects customers • Identify and form relationships with "brand influencers" • Gauge employee morale by analyzing internal postings • Discover important trends by analyzing customer responses • Identify specific causes for brand decline, such as long wait times • Identify emotion conveyed in voices and faces

FIGURE 7.32 5 AI technologies driving business value

Besides these applications, speech recognition software is also used for the following:

Giving special attention to certain words, such as those spoken frequently or that are prominent in a conversation.

A speech recognition software with acoustic training can be used to differentiate between speaking style, pace and volume of people speaking in an office to understand the contributions of individuals in a conversation.

The software can also be used for filtering profanity that filters out undesirable words and language. Other applications of NLP and speech recog-

nition in the industry are given in **Fig. 7.32**

7.16.3 Algorithms for Speech Recognition

Speech recognition has been possible due to some powerful algorithms listed below.

Natural language processing. eases and accelerates the speech recognition process.

1. ***Hidden Markov model:*** This is used in autonomous systems in two cases. First, the state is partially observable. Second, when the entire information required for decision-making is not immediately available to the sensor or the microphone for speech recognition.
2. ***N-grams.*** This technique creates a probability distribution for a sequence. Such an algorithm looks at the last few words spoken, approximates the history of the sample of speech to determine the probability of the next word that will be spoken.

Artificial intelligence, NLP and machine learning/deep learning techniques are used in advanced speech recognition software. These systems learn the grammar, structure, syntax and composition of audio and voice signals to process speech. These systems learn with each use to give better results the next time.

7.16.4 Advantages of Speech Recognition

1. Speech recognition facilitates machine-to-human communication in natural language.
2. The technology can be easily used in computers and mobile devices, making it readily accessible.
3. Even a novice user can easily use speech recognition feature that is straightforward to operate and often runs in the background.
4. Speech recognition systems that use AI techniques automatically improve with every use as more and more data is generated. So, these

systems evolve continuously to become more effective and easier to use over time.

7.16.5 Disadvantages of Speech Recognition

1. Speech recognition systems often give inconsistent performance as they may not always capture words accurately possibly due to difference in pronunciation, lack of support for a particular language and background noise. Although acoustic training does help, it is however not perfect.
2. These systems take time to deploy and master and are relatively slow.

The success of speech recognition system also depends on the recording equipment used and not just the software alone.



PROGRAMMER'S ZONE

Program for sentence tokenization on a review written for the Jruma app.

```
import nltk
text = "A very great App for students! I wish to solve more and
more quizzes every day and increase my knowledge! Thank you so
much for this App! This App should be appreciated more and more!
Congratulations!!:))"
sentences = nltk.sent_tokenize(text)
for sentence in sentences:
    print(sentence)
```

OUTPUT

```
A very great App for students!
I wish to solve more and more quizzes every day and increase my knowledge!
Thank you so much for this App!
This App should be appreciated more and more!
Congratulations!!
:))
```

Write a script to tokenize words

```
from nltk import word_tokenize, sent_tokenize
import nltk
text = "A very great App for students! I wish to solve more and
more quizzes every day and increase my knowledge! Thank you so
much for this App! This App should be appreciated more and more!
Congratulations!!:))"
words = nltk.word_tokenize(text)
for word in words:
    print(word, end = '###')
```

OUTPUT

```
A###very###great###App###for###students###!###I###wish###to
###solve###more###and###more###quizzes###every###day###and
###increase###my###knowledge###!###Thank###you###so###much
###for###this###App###!###This###App###should###be###
appreciated###more###and###more###!###Congratulations
###!###!###:###)###)###
```

Program to demonstrate stemming text.

```
from nltk.stem import PorterStemmer
stemmer = PorterStemmer()
words = ['run', 'running', 'runs', 'runner']
stemmed = [stemmer.stem(word) for word in words]
print(stemmed)
```

OUTPUT

```
['run', 'run', 'run', 'runner']
```

Remove stop words from our text

```
Note that you must first download stop words of English language by
writing nltk.download('stopwords')
from nltk import word_tokenize, sent_tokenize
from nltk.corpus import stopwords
import nltk
nltk.download('stopwords')
text = "A very great App for students! I wish to solve more and
more quizzes every day and increase my knowledge! Thank you so
much for this App! This App should be appreciated more and more!
Congratulations!!:))"
stop_words = stopwords.words('english')
words = nltk.word_tokenize(text)
cleaned_text = []
for word in words:
    if word not in stop_words:
        cleaned_text.append(word)
print("Uncleaned Text : ", text)
print("Cleaned Text : ", cleaned_text)
```

OUTPUT

```
Uncleaned Text :  A very great App for students! I wish to solve
more and more quizzes every day and increase my knowledge! Thank you
so much for this App! This App should be appreciated more and more!
Congratulations!!:))
Cleaned Text :  ['A', 'great', 'App', 'students', '!', 'I', 'wish',
'solve', 'quizzes', 'every', 'day', 'increase', 'knowledge', '!',
'Thank', 'much', 'App', '!', 'This', 'App', 'appreciated', '!',
'Congratulations', '!', '!', ':', ')', ')']
```

Program to tag text using the pos_tag() method of nltk library.

```
Note you need to first download the averaged_perceptron_tagger'
before using the po_tag() method
from nltk import word_tokenize, sent_tokenize
from nltk.corpus import stopwords
import nltk
nltk.download('averaged_perceptron_tagger')
text = "A very great App for students! I wish to solve more and
more quizzes every day and increase my knowledge! Thank you so
much for this App! This App should be appreciated more and more!
Congratulations!!:))"
stop_words = stopwords.words('english')
words = nltk.word_tokenize(text)
cleaned_text = []
for word in words:
    if word not in stop_words:
        cleaned_text.append(word)
tagged = nltk.pos_tag(cleaned_text)
print("Tagged Text is : ", tagged)
```

OUTPUT

```
Tagged Text is :  [('A', 'DT'), ('great', 'JJ'), ('App', 'JJ'),
('students', 'NNS'), ('!', '.'), ('I', 'PRP'), ('wish', 'VBP'),
('solve', 'RB'), ('quizzes', 'JJ'), ('every', 'DT'), ('day', 'NN'),
('increase', 'VB'), ('knowledge', 'NN'), ('!', '.'), ('Thank', 'NNP'),
('much', 'JJ'), ('App', 'NNP'), ('!', '.'), ('This', 'DT'), ('App',
'NNP'), ('appreciated', 'VBD'), ('!', '.'), ('Congratulations',
'NNPS'), ('!', '.'), ('!', '.'), (':', ':'), (')', ')'), (')', ')')]
 In the output, VB refers to 'verb', NNS refers to 'plural nouns', DT
 refers to a 'determiner'.
```

Write a code to compare the working of a lemmatizer and a stemmer.

```
from nltk.stem import PorterStemmer, WordNetLemmatizer
from nltk.corpus import wordnet
import nltk
nltk.download('wordnet')
def compare_stemmer_and_lemmatizer(stemmer, lemmatizer, word, pos):
    print("Stemmer:", stemmer.stem(word))
    print("Lemmatizer:", lemmatizer.lemmatize(word, pos))
lemmatizer = WordNetLemmatizer()
stemmer = PorterStemmer()
compare_stemmer_and_lemmatizer(stemmer, lemmatizer, word = "ran",
pos = wordnet.VERB)
#compare_stemmer_and_lemmatizer(stemmer, lemmatizer, word = "drove",
pos = wordnet.VERB)
```

OUTPUT

```
Stemmer: ran
Lemmatizer: run
```

Write a code to print the two most frequently occurring words in a text.

```
message = ['She', 'sells', 'sea', 'shells', 'on', 'the',
          'sea', 'shore', 'she','said', 'on', 'the', 'sea', 'shore']
tokens = set(message)
tokens = sorted(tokens)
print(tokens[-2:])
OUTPUT
['shore', 'the']
Write a code to print the 10 most frequently occurring character in
a piece of text.
import nltk
text1 = '''
Computer Network Types
A computer network is a group of computers linked to each other that
enables the computer to communicate with another computer and share
their resources, data, and applications.
A computer network can be categorized by their size. A computer
network is mainly of four types:
Computer Network Types
LAN(Local Area Network)
PAN(Personal Area Network)
MAN(Metropolitan Area Network)
WAN(Wide Area Network)'''
fdist1 = nltk.FreqDist(text1)
print(fdist1.most_common(10))
```

OUTPUT

```
[(' ', 55), ('e', 44), ('t', 34), ('o', 33), ('r', 33), ('a', 23),
('p', 15), ('c', 14), ('n', 14), ('i', 14)]
```

Write a code to print words in a text that have more than 7 characters.

```
message = 'A computer network is a group of computers linked to each
other that enables the computer to communicate with another computer
and share their resources, data, and applications. A computer network
can be categorized by their size. A computer network is mainly of
four types'
mes = message.split(' ')
long_words = [w for w in mes if len(w) > 7]
print(sorted(long_words))
```

OUTPUT

```
['applications.',    'categorized',    'communicate',    'computer',
'computer',   'computer',   'computer',   'computer',   'computers',
'resources,']
```

Create bigrams using a piece of quotation.

```
import nltk
text = 'All our dreams can come true, if we have the courage to
pursue them.'
text = text.split(' ')
print(list(nltk.bigrams(text)))
```

OUTPUT

```
[('All', 'our'), ('our', 'dreams'), ('dreams', 'can'), ('can',
'come'), ('come', 'true,'), ('true,', 'if'), ('if', 'we'), ('we',
'have'), ('have', 'the'), ('the', 'courage'), ('courage', 'to'),
('to', 'pursue'), ('pursue', 'them.')]
```

Print collocations from the given text.

Hint: A collocation is a sequence of words that occur together unusually often

```
import nltk
message = 'Information technology (IT) refers to everything that
businesses use computers for. Information technology is building
communications networks for a company, safeguarding data and
information, creating and administering databases, helping employees
troubleshoot problems with their computers or mobile devices, or
doing a range of other work to ensure the efficiency and security of
business information systems. Demand for professionals in this field
is high and growing, and people entering the field have a range of
career paths to choose from.'
tokens = nltk.word_tokenize(message)
text = nltk.Text(tokens)
print(text.collocations())
```

OUTPUT

```
Information technology
```

Write a program that reads a piece of text and then returns how frequent the different lengths of word are.

```
import nltk
message = '''There are other skills and expertise that are highly
desirable as well, but these are the primary four in my opinion.
These will be referred to as the data scientist pillars for the rest
of this article.
In reality, people are often strong in one or two of these pillars,
but usually not equally strong in all four. If you do happen to
meet a data scientist that is truly an expert in all, then you've
essentially found yourself a unicorn.
Based on these pillars, my data scientist definition is a person
who should be able to leverage existing data sources, and create
new ones as needed in order to extract meaningful information and
actionable insights. A data scientist does this through business
domain expertise, effective communication and results interpretation,
and utilization of any and all relevant statistical techniques,
programming languages, software packages and libraries, and data
infrastructure. The insights that data scientists uncover should be
used to drive business decisions and take actions intended to achieve
business goals.
'''
tokens = nltk.word_tokenize(message)
text = nltk.Text(tokens)
fdist = nltk.FreqDist(len(w) for w in text)
print(fdist)
print(fdist.most_common)
```

OUTPUT

```
<FreqDist with 13 samples and 189 outcomes>
<bound method Counter.most_common of FreqDist({3: 31, 2: 31, 4: 24,
1: 23, 7: 17, 5: 13, 6: 13, 8: 13, 9: 11, 11: 5, ...})>
```

Remarks: In the above program, the FreqDist function is used to analyze the distribution of word lengths in a text. In its output, each number is the length of the corresponding word in the text. From the above output, it is clear that majority of the words are of 3 and 2 characters long. There are 24 words of length 4 and 23 words of length 1, so on and so forth

Example: Implementing Genetic Algorithm

```
import numpy
data_inputs = numpy.array([[1, 1],
                           [1, 0],
                           [0, 1],
                           [0, 0]])
daat_outputs = numpy.array([0,
                            1,
                            1,
                            0])
num_neurons_input = 2
num_neurons_output = 2
num_neurons_hidden_layers = [2]
hidden_activations=["relu"]
output_activation = "softmax"
num_solutions = 5
import pygad.gann
GANN_instance = pygad.gann.GANN(num_solutions=5,
                                num_neurons_input=2,
                                num_neurons_hidden_layers=[2],
                                num_neurons_output=1,
                                hidden_activations=["relu"],
                                output_activation="softmax")
#print(GANN_instance.population_networks)
population_vectors  =  pygad.gann.population_as_vectors(population_
networks=GANN_instance.population_networks)
print(population_vectors)
```

OUTPUT

```
[array([-0.08967126,  -0.04649616,     0.08264154,     0.00373608,
-0.04776243,
        0.00836679]), array([ 0.07586315, -0.00324067,  0.04323463,
0.07810089, -0.07372201,
        0.05216289]), array([-0.09859114, -0.07508463, -0.03051525,
0.00402294, -0.03516184,
        0.06906154]), array([ 0.08227591,  0.04856971, -0.0100997 ,
-0.02059963, -0.04185758,
       -0.01908773]), array([ 0.00494026, -0.05366674,  0.05792332,
-0.06749663,  0.03835708,
       -0.06406028])]
```



KEY TERMS

Natural language processing: The manipulation or understanding of text or speech by a computer software.

Chatbots: NLP, AI-based systems that are specifically designed to interact with humans through text or speech.

Lexical analysis: The process of identifying and analysing structure of words by breaking down a big chunk of text into elementary words.

Syntactic analysis: The process of parsing a sentence to analyse words in the sentence to interpret its meaning as per the grammatical rules of the language.

Parts of speech (POS) tagging: The process that marks words in text format for a particular part of a speech depending on its definition and context.

Syntax: Arrangement of words in a sentence to make grammatical sense.

Perception: The ability to see, hear or become aware of something through senses.

Context-free grammar: A formal grammar that can be used to generate all possible strings in a given formal language.

Derivation: A sequence of rule applications that derive a terminal string $w = w_1 \ldots w_n$ from S.

Parse tree: A graphical representation of terminal as well as non-terminal symbols.

Parser: A compiler that breaks the data coming from lexical analysis phase into smaller elements.

Speech recognition: The ability of a machine to identify words spoken aloud to convert them into readable text.



RECAPITULATION POINTS

- NLP can be used to create intelligent systems for speech recognition, machine translation, question answering, spam detection, document summarization, predictive typing, named entity recognition and autocomplete, to name a few.

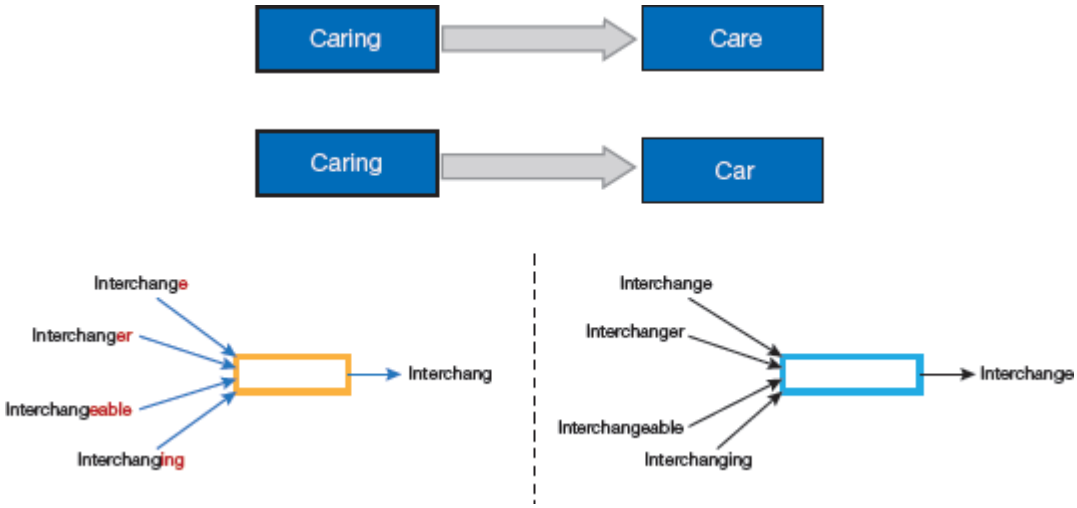
- While writing using human language, we often misspell or abbreviate words, or omit punctuation. Similarly, while speaking, we pronounce different words differently (in regional accents). Some people mumble, stutter and borrow terms from other languages which further increases the complexity of processing natural languages.
- Techniques like deep, supervised and unsupervised learning are widely used for modelling human language to resolve ambiguity in language.
- NLP systems break down text/speech in human language into shorter, elemental pieces, to understand relationships that exist between the pieces for interpreting its meaning.
- Speech recognition is used for converting spoken words into text.
- Topic classification uses NLP to organize unstructured text into categories.
- Natural language understanding (NLU) deals with mapping the given input in natural language into useful representations.
- Natural language generation (NLG) uses content in internal representation to produce meaningful sentences in a natural language.
- A chatbot works on request-response model.
- The lexicon of a language refers to the collection of words and phrases in that language.
- Discourse integration studies the meaning of a sentence with respect to the meaning of sentence preceding as well as succeeding it in sequence.
- In word sense disambiguation step, the meaning of a word is analysed based on the context in which it is used.
- Named entity recognition identifies words that can be classified in different groups.
- Chatbots provide automated answers to common customer queries. They use pattern recognition systems with heuristic responses to interact with users.
- Lexical ambiguity is the most primitive level of ambiguity that exists at the level of individual words. It arises when one or more words in a sentence has multiple meanings.
- *The recency rule believes that in 85% of the cases, the referent is the most recently mentioned object of correct gender and number.*

- A context-free grammar G can be defined using a 4-tuple (N, T, S, P) value, where, N is the set of non-terminal symbols or variables, T is the set of terminal symbols, S refers to the Start symbol and P is the set of Production rules. Every production rule consists of non-terminals, arrow and terminals.
- In a production rule, the left side of production is non-terminals while the right side is called terminals.
- *A grammar is said to be ambiguous if there exists more than one leftmost derivation or more than one rightmost derivative or more than one parse tree for a given input string.*
- N-grams is an algorithm for speech recognition that creates a probability distribution for a sequence. It looks at the last few words spoken to determine the probability of the next word that will be spoken.

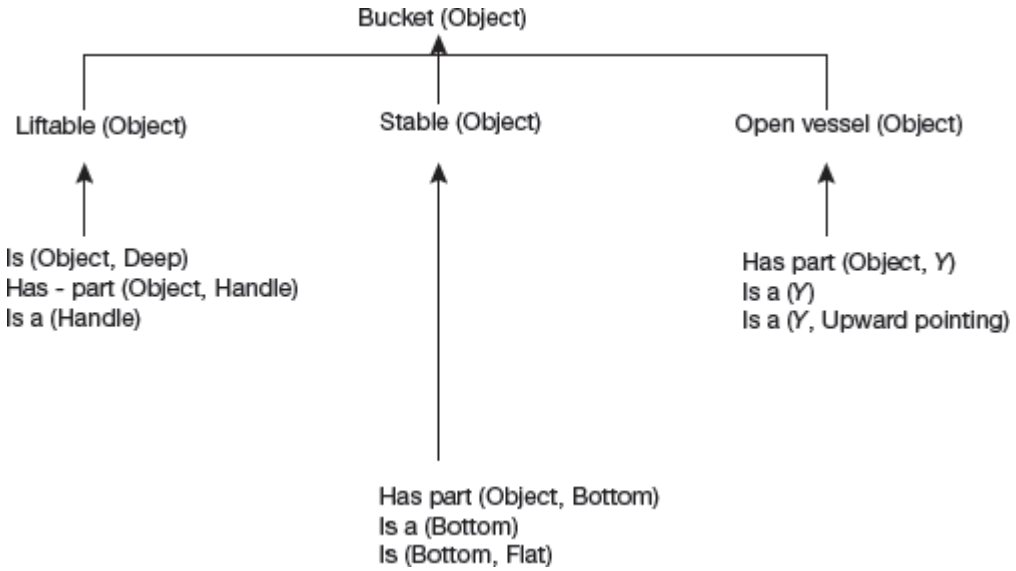


REVIEW QUESTIONS

1. What do you understand by the term natural language processing?
2. Give the applications of NLP.
3. Discuss some issues faced while dealing with human language.
4. How are NLP systems useful?
5. Give any three real-world applications of NLP.
6. Discuss three ambiguities faced by NLP systems.
7. Discuss the steps involved to perform NLP.
8. What is a chatbot? Give any three uses of a chatbot.
9. How does a chatbot work?
10. Enumerate the different types of chatbots.
11. What steps will you follow to create a chatbot?
12. Can you identify which is stemming and which one of the two is lemmatization?



13. Differentiate between the following.
    1. Natural language and computer language.
    2. NLU and NLG
    3. Open and closed chatbot
    4. Lemmatization and stemming
    5. Syntax versus semantic analysis
    6. Leftmost derivation and rightmost derivation
    7. Top-down and bottom-up parsing
    8. Speech and voice recognition
14. Summarize the five steps involved in any NLP system.
15. Discuss any five applications of NLP systems.
16. Highlight the problems faced when implementing NLP systems.
17. Give some techniques to resolve ambiguity.
18. What is ambiguous grammar? Draw a parse tree for such a grammar.
19. How does speech recognition work?
20. Write short notes on:
    1. Natural language processing
    2. Context-free grammar
    3. Genetic algorithms
21. Draw the tree that represents the sentence, ‘This movie is revealing the exact incidence’.
22. Consider the structure given below that describes a bucket. Write some rules that you can derive from this structure.



23. Why is the sentence, ‘Mary ate a salad with spinach from California for lunch on Tuesday.’, syntactically ambiguous?
24. Create grammar to parse the sentence,
    1. ‘The bird pecks the grains’
    2. ‘The dog saw a man in the house’
25. Why is the sentence, ‘John kissed his wife, and so did Sam.’ semantically ambiguous?
26. Find out, who is being referred to by pronouns.
    1. ‘Margaret invited Susan for a visit, but she told her she had to go to work’.
    2. ‘Margaret invited Susan for a visit, and she gave her a good lunch.’
    3. ‘On the train to Boston, George chatted with another passenger. The man turned out to be a professional hockey player.’
    4. ‘Bill told Amy that he had decided to spend a year in Italy to study art.’ ‘That would be his life’s work.’
27. Why is the sentence, ‘Kiran went to Sunita. She said, ‘I am hungry.’ said to be an example of referential ambiguity?
28. ‘I saw the girl with binoculars’, is syntactically ambiguous. Comment.
29. ‘Elsie tried to reach her aunt on the phone, but she didn’t answer.’ is lexically as well as syntactically ambiguous. Why? Is there any other ambiguity also involved?
30. Identify the issue- (ellipsis, associational frequency, semantic constraints):
    1. ‘Mozart was born in Salzburg and Beethoven, in Bonn’.
    2. Identify the resolution ‘I hammer the nail and I saw the board’.
    3. ‘The sick bat lay on the ground.’

4. ‘The clock is fast.’ vs ‘The horse is fast.’ vs. ‘The clothes are fast.’ vs. ‘The slopes are fast.’ vs. ‘The knot is fast.’
5. ‘Manya is looking for a **match**.’, is lexically ambiguous. Comment.



FILL UP

1. Google uses _____ to detect and filter spam emails.
2. In NLP, text is broken down into smaller units called _____.
3. _____ identifies words that can be classified in different groups.
4. In _____, algorithms are used to analyse unstructured data and automatically generate content based on that data.
5. _____ provide automated answers to common customer queries.
6. _____ ambiguity usually happens when pronouns are used.
7. _____ occurs when we omit words that are actually needed for grammatical completion of the syntax.
8. _____ uses the most plausible interpretation is used.
9. _____ is a superset of regular grammar.
10. _____ is a compiler that breaks the data coming from lexical analysis phase into smaller elements.
11. NLP applies machine learning algorithms to _____ and _____.
12. _____ uses NLP to detect and filter spam emails.
13. _____ language is ambiguous in nature.
14. _____ analysis is used to identify user’s opinion within large amounts of text.
15. Speech _____ is used for converting spoken words into text.
16. _____ are NLP-based systems that are specifically designed to interact with humans through text or speech.
17. _____ of a language refers to the collection of words in a language.
18. _____ analysis is done to analyse words in the sentence to interpret its meaning as per the grammatical rules of the language.
19. _____ reduces words to their base form.
20. *Words like I, they, have, like, yours* are known as _____ words.
21. _____ refers to having any sentence depends upon the meaning of the sentence just preceding it.
22. NLP and machine learning are both subsets of _____.
23. A chatbot works on _____ model.



STATE TRUE OR FALSE

1. An NLP system takes a language as input only in the spoken form.
2. In semantic analysis, what was said is re-interpreted on what it actually meant.
3. *Syntax* involves the use of and meaning behind words.
4. In named entity recognition step, the meaning of a word is analysed based on the context in which it is used.
5. Syntax level ambiguity arises when one or more words in a sentence have multiple meanings.
6. *Textual coherence believes that in 85% of the cases, the referent is the most recently mentioned object of correct gender and number.*
7. Every production rule consists of non-terminals, arrow and terminals.
8. In a production rule, the right side of production is non-terminals while the left side is called terminals.
9. In context-free grammar, parse tree follows the precedence of operators.
10. Voice recognition is the ability of a machine to identify words spoken aloud to convert them into readable text.
11. NLP applies machine learning algorithms to process text and speech.
12. Predictive typing uses computer vision algorithms to recommend what users should type next.
13. Siri is the virtual assistant of Google.
14. Every language has its unique set of grammar and syntax rules, terms and slang.
15. Computer language has a very large vocabulary.
16. Computer language is easily understood by humans.
17. Chatbots are used to offer 24 X 7 assistance to customers by handling their queries.
18. Chatbots are used to automate repetitive and time-consuming tasks.
19. Natural language understanding generates non-linguistic outputs from natural language inputs.
20. NLU is harder than NLG.
21. Syntax level ambiguity is common when using pronouns.
22. Stop-word removal process labels tokens as *verb, adverb, adjective, noun*, etc.

23. Word tokenization divides a string written in a language into its basic components.
24. Open chatbot executes a specific conversation flow.



MULTIPLE CHOICE QUESTIONS

1. _____ of a language refers to the collection of words and phrases in that language
    1. Syntax
    2. Semantics
    3. Lexicon
    4. All of these
2. _____ studies the meaning of a sentence with respect to meaning of sentence preceding as well as succeeding it in sequence.
    1. Lexicon analysis
    2. Syntax analysis
    3. Semantic analysis
    4. Discourse integration
3. _____ is an arrangement of words in a sentence to make grammatical sense.
    1. Syntax
    2. Semantics
    3. Lexicon
    4. All of these
4. During _____ analysis, processes like tokenization, stemming, lemmatization, stop words removal, POS tagging are performed.
    1. Lexicon
    2. Syntactic
    3. Semantic
    4. Pragmatic
5. Natural language _____ uses a database to determine semantics behind words and generate new text.
    1. Understanding
    2. Processing
    3. Generation
    4. Analysis

6. _____ helps to automatically summarize text and search important pieces of data.
    1. Text classification
    2. Text extraction
    3. Text generation
    4. Machine translation
7. _____ ambiguity occurs when a sentence can be parsed in different ways.
    1. Lexical
    2. Syntax level
    3. Semantic
    4. Referential
8. _____ maps a sentence plan into a sentence structure.
    1. Text planning
    2. Sentence planning
    3. Text realization
    4. Ellipsis
9. In parsing a tree, _____ traversal method is used.
    1. Pre-order
    2. Post-order
    3. In-order
    4. Any of these
10. A grammar is said to be ambiguous if there exists _____.
    1. more than one leftmost derivation
    2. more than one rightmost derivative
    3. more than one parse tree for a given input string
    4. any of these
11. _____ is the manipulation or understanding of text or speech by a computer software.
    1. Computer vision
    2. Natural language processing
    3. Robotics
    4. Machine learning
12. NLP systems _____ to user queries.
    1. Interact
    2. Understand
    3. Respond

4. All of these

13. _____ is the virtual assistant of Microsoft.
    1. Alexa
    2. Siri
    3. Cortona
    4. Ymo
14. Sentiment analysis is used to assess the _____ of the customers.
    1. Attitude
    2. Behaviour
    3. Emotional state
    4. All of these
15. NLP systems cannot be used for _____.
    1. Predictive typing
    2. Spelling correction
    3. Speech translation
    4. None of these
16. _____ uses NLP to organize unstructured text into categories.
    1. Document summarization
    2. Topic classification
    3. Predictive typing
    4. Speech translation
17. _____ use NLP to analyse hundreds of open-ended responses to surveys to gain insights from customer feedback.
    1. Document summarization
    2. Topic classification
    3. Predictive typing
    4. Speech translation
18. _____ ambiguity exists due to confusion in different words.
    1. Lexical
    2. Syntax level
    3. Referential
    4. Grammatical
19. ---------- ambiguity is common when using pronouns.
    1. Lexical
    2. Syntax level
    3. Referential
    4. Grammatical

20. _____ analysis is the process of identifying and analysing the structure of words by breaking down a big chunk of text into elementary words.
    1. Lexical
    2. Syntactic
    3. Semantic
    4. Pragmatic
21. _____ means breaking a text into elementary words.
    1. Part of speech tagging
    2. Tokenization
    3. Lemmatization and stemming
    4. Stop-word removal
22. Semantic analysis draws the exact meaning from the text.
    1. Lexical
    2. Syntactic
    3. Semantic
    4. Pragmatic
23. In _____ step, text is re-interpreted based on what it actually meant in the current context.
    1. Lexical
    2. Syntactic
    3. Semantic
    4. Pragmatic
24. _____ chatbots follow predefined paths while conversing with the user.
    1. Rule-based
    2. Menu-based
    3. Intellectually-independent
    4. AI-powered



ANSWERS

Fill Up

1. Natural language processing
2. Tokens

3. Named entity recognition
4. Natural language generation
5. Chatbots
6. Referential
7. Ellipsis
8. *World knowledge*
9. Context-free grammar
10. Parser
11. text, speech
12. Gmail
13. Natural
14. sentiment
15. Translation
16. Chatbots
17. Lexicon
18. Syntactic
19. Lemmatization and stemming
20. stop
21. Discourse Integration
22. Artificial intelligence
23. request-response

State True or False

1. False
2. False
3. False
4. False
5. False
6. *False*
7. True
8. False
9. True
10. False
11. True
12. False
13. False

14. True
15. False
16. False
17. True
18. True
19. False
20. True
21. False
22. False
23. False
24. False

Multiple Choice Questions

1. c
2. d
3. a
4. b
5. c
6. b
7. b
8. c
9. c
10. d
11. b
12. d
13. c
14. d
15. d
16. b
17. b
18. a
19. c
20. a
21. b
22. c
23. d

24. a